



Contents lists available at ScienceDirect

Journal of Corporate Finance

journal homepage: www.elsevier.com/locate/jcorpfin

Dividends and economic policy uncertainty: International evidence[☆]

Najah Attig^a, Sadok El Ghouli^b, Omrane Guedhami^c, Xiaolan Zheng^{d,*}

^a Sobey School of Business Saint Mary's University, Halifax, Nova Scotia B3H 3C3, Canada

^b Campus Saint-Jean, University of Alberta, 8406, Rue Marie-Anne-Gaboury (91 Street), Edmonton, AB T6C 4G9, Canada

^c Moore School of Business, University of South Carolina, 1014 Greene Street, Columbia, SC 29208, USA

^d Nottingham University Business School China, University of Nottingham Ningbo China, 199 Taikang East Road, Ningbo 315100, China

ARTICLE INFO

Keywords:

Dividends

Economic policy uncertainty

Agency theory

Corporate governance

International

ABSTRACT

We provide the first international evidence on the impact of economic policy uncertainty (EPU) on dividend policy. Using data from 19 countries, we find that a high level of EPU is positively associated with dividend payout. This evidence is robust to using alternative dividend payout measures, to controlling for other sources of uncertainty, and to addressing endogeneity. We further find that the effect of EPU on dividend policy is moderated by firms' free cash flows and governance quality, and by the quality of country-level indicators of shareholder protection, disclosure, enforcement, and creditor protection. Collectively, our novel evidence suggests that dividends help mitigate agency problems during high-EPU periods.

1. Introduction

Research on the importance of uncertainty for dividend policy has gained momentum in recent years in light of the 2008–2009 financial crisis and subsequent global economic downturn. This body of work includes Abreu and Gulamhussein (2013), Bliss et al. (2015), and Attig et al. (2016), who show that firms adjust their dividend policies in response to the crisis. More recently, Farooq and Ahmed (2019) find that firms pay higher dividends during presidential election years than non-election years. Much of the related evidence, however, is based on event-driven uncertainty (e.g., political elections, financial crisis) and thus prone to the criticism of overlooking changes in uncertainty between non-event periods (Brogaard and Detzel 2015; Matousek et al. 2020). Our study fills this important void in the literature by providing the first cross-country evidence of the impact of economic policy uncertainty (EPU) on dividend policy.

EPU refers broadly to uncertainty about government actions that affect the economic environment. While policy uncertainty is unobservable and can be difficult to quantify, Baker et al. (2016) (henceforth BBD) use newspaper coverage frequency to capture

[☆] The authors acknowledge with much appreciation the comments and suggestions by Bart Lambrecht (Editor), two anonymous reviewers, Itzhak Ben-David, Narjess Boubakri, Ruiyuan Chen, Sattar Mansi, David Moore (discussant), He Wang and participants at the 2018 Financial Management Association Meeting and Dalhousie University's Mackay Finance Seminars for their helpful comments and suggestions. Najah Attig and Sadok El Ghouli appreciate the generous financial support from Canada's Social Sciences and Humanities Research Council (435-2013-2098, 435-2020-1156).

* Corresponding author.

E-mail addresses: najah.attig@smu.ca (N. Attig), elghoul@ualberta.ca (S. El Ghouli), omrane.guedhami@moore.sc.edu (O. Guedhami), xiaolan.zheng@nottingham.edu.cn (X. Zheng).

<https://doi.org/10.1016/j.jcorpfin.2020.101785>

Received 5 March 2019; Received in revised form 28 October 2020; Accepted 12 November 2020

Available online 21 November 2020

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uncertainty about “*who* will make economic policy decisions, *what* economic policy actions will be undertaken and *when*, and the economic effects of policy actions (or inaction)” (p. 1598). Their measure requires that a newspaper article cover terms relevant to the three categories of uncertainty, the economy, and policy, and capture both near- and longer-term concerns.¹

Investigating the impact of EPU on dividend policy represents an important distinction from prior literature. EPU is different from other types of uncertainty. First, EPU differs from macroeconomic uncertainty, which relates to uncertainty about macroeconomic fundamentals, as EPU refers to uncertainty about government actions that affect the economic environment (Beckmann and Czudaj 2017). Second, it differs from event-related uncertainty, such as political elections and financial crises, as it quantifies uncertainty outside the timeframe captured by these events (Brogaard and Detzel 2015), and thus is more persistent than event-driven uncertainty.² Third, EPU differs from firm-level uncertainty as it results from government policy changes and regulatory shocks, and from other shocks that are similarly beyond a manager’s control (e.g., wars, trade sanctions). This makes EPU largely exogenous and more difficult to hedge for firms and investors than firm-level uncertainty (Duong et al. 2020; El Ghouli et al. 2020). Finally, EPU is particularly relevant for the purpose of our multinational study because it reflects a wide and varying mix of events and concerns across countries and time (Davis 2017) and varies over time (BBD, 2016), which may impact investors’ time-varying risk preferences and concerns about managerial self-dealing, hence their demands for dividends.

Anecdotal evidence suggests, however, that despite the unusually high level of policy uncertainty in recent years (BBD, 2016), some firms have increased their dividends to record levels. For example, in 2012, ExxonMobil increased its quarterly dividend by 21% to become the world’s largest corporate dividend payer. Also in 2012, Cisco Systems Inc. raised its quarterly dividend by 75%, and Apple paid its first dividend since 1995. More broadly, the Janus Henderson Global Dividend Index³ (2018) suggests that firms in the U.S., Canada, Japan, and other markets such as emerging markets, Europe, and Asia Pacific, notched up record dividend payouts “despite fears over global economic growth and Brexit-related uncertainty” (Beioley 2019). On the surface, these changes appear counterintuitive because precautionary motives suggest that, in response to heightened uncertainty, firms should retain a greater portion of their earnings, and hence decrease dividend payments, to buffer investments from financing shocks.

The agency-theoretic view, however, would suggest a positive association between EPU and dividends. Contrary to Miller and Modigliani’s (1961) payout policy irrelevance (in a world of perfect information) and in the absence of complete contracts, subsequent theoretical studies suggest that dividends can play an important role in mitigating agency problems between corporate insiders and outside shareholders (e.g., Easterbrook 1984; Jensen 1986; Fluck 1999; Myers 2000; Zwiebel 1996). These theories suggest that shareholders prefer dividends over retained earnings. The surveys by Allen and Michaely (2003) and Leary and Michaely (2011) indicate that accumulated evidence appears to be consistent with the role of dividends in mitigating agency problems. Farre-Mensa et al.’s (2014) survey suggests also that many of the potential conflicts of interest within the firm interact with payout policy. For instance, paying out dividends mitigates type I agency problems in widely held firms (e.g., Rozeff 1982; DeAngelo et al. 2006; Lippert et al. 2000) and type II agency problems in closely held firms (e.g., Faccio et al. 2001; Pindado et al. 2012; Berzins et al. 2018). Importantly, La Porta et al. (2000) conclude that agency theory is highly relevant in explaining dividend policies around the world.

Of relevance to our focus, and drawing on the insights that effective corporate governance can mitigate the stronger incentives of insiders to divert corporate resources during bad times (e.g., Bae et al. 2012; Baek et al. 2004; Johnson et al. 2000; Mitton 2002), we posit that investors’ preferences for dividends, as a means to address the agency problems of free cash flows, will be more pronounced during periods of heightened EPU. Firms are likely to be more vulnerable to agency problems during high-EPU periods because increased EPU has negative implications for various macroeconomic activities (BBD, 2016) and for corporate capital investment (Gulen and Ion 2016).⁴ This in turn will increase cash reserves, which can be easily turned into private benefits (Myers and Rajan 1998). As such, higher payouts during increased EPU periods can make managers more disciplined in the use of the firm’s resources. In this context, managers will tap into capital markets to fund future investments (e.g., Easterbrook 1984), which will expose them to subsequent monitoring by the market.

The studies of Floyd et al. (2015) and Abreu and Gulamhussen (2013) are particularly germane to the focus of our paper. Floyd et al. (2015) compare the payout policies of U.S. industrials and banks over the past 30 years, show that dividends are resilient, and conclude that, for industrials, the free cash flow explanation is the most important reason dividends survive. Abreu and Gulamhussen (2013) examine dividend payouts of U.S. bank holding companies around the global financial crisis and conclude that the agency cost hypothesis explains dividend payouts before and during (more pronouncedly) the financial crisis and that more dividends are paid to accommodate the increased need for monitoring. Other studies lend further credence to the agency explanation of dividends. For instance, Officer (2011) shows that firms with “good governance” pay lower levels of dividends, while Grinstein and Michaely (2005)

¹ Examples of policy-relevant terms include, “Congress,” “deficit,” “Federal Reserve,” “legislation,” “regulation,” or “White House,” in the newspaper’s native language. Differences in the supervising agency’s name (e.g., “Bank of Japan” for Japan), as well as terms specific to a nation (e.g., “customs duties” for India) are accounted for, as well as abbreviations and term variants, such as “uncertainties” and “regulatory.” The next section provides a detailed discussion of Baker et al.’s (2016) index.

² Prior research on EPU shows that its effects on firms and stock markets extend beyond other measures of uncertainty (Bonaime et al. 2018; Brogaard and Detzel 2015; Gulen and Ion 2016). To isolate the incremental effect of EPU on dividends in our analysis, we control for various sources of uncertainty (e.g., election-, firm-, industry-, and country-level macroeconomic uncertainty).

³ The Janus Henderson Global Dividend Index is a long-term study of global dividend trends. It analyzes quarterly dividends paid by the 1200 largest firms by market capitalization.

⁴ Recent empirical literature shows that policy-induced uncertainty depresses real economic activities (e.g., BBD, 2016), decreases firm-level capital investment (Gulen and Ion 2016), reduces capital flows, and limits economic recovery (Bonaime et al. 2018).

document that firms with high institutional holdings tend to pay lower dividends. Studying Asian and European firms, [Faccio et al. \(2001\)](#) find that firms that are vulnerable to expropriation (closely affiliated with a business group and lower ownership-to-control ratio) pay significantly higher dividends.

In this study, we assume that the direction of the link between EPU and dividends is an open empirical question. Using a sample of 190,799 firm-year observations for 23,042 firms from 19 countries over 1991–2015, we find that EPU has a consistently positive and significant effect on dividends as measured by the dividend payout ratio, the dividend yield, the ratio of cash dividends to total sales (cash flows), and the total payout ratio. Our main evidence remains valid when we use change regressions in place of level regressions (in the spirit of [Faccio and Xu 2015](#)).

A skeptical view, however, might argue that this evidence is merely picking up the adverse effects of policy uncertainty on the denominators of our key variables (sales, net income, market capitalization), and is not a genuine response to such an uncertainty. To rule out this alternative interpretation, we examine the effects of EPU on the denominators of our payout variables, and find no significant effects. We further replicate our analysis using cash dividend per share to common stockholders (DPS) and DPS for dividend payer. We continue to estimate a positive and significant effect of EPU on these additional measures of dividend payout. Importantly, this evidence remains valid when we examine the impact of EPU on the cumulative abnormal return around the announcement of dividend increase.⁵

Similar to other EPU studies, one concern is that our results may reflect other sources of uncertainty, and not necessarily policy uncertainty. We address this concern by accounting for other sources of uncertainty. Controlling for uncertainty surrounding political elections, we find that our main results remain unchanged, suggesting that the effect of EPU on dividends captures the effect of aggregate economic policy-induced uncertainty that goes beyond the impact of political election uncertainty. Our results also hold when we control for firm-, industry-, and country-level macroeconomic uncertainty, which suggests that the effect of EPU on dividend payout policy is distinct from that of other sources of uncertainty.

Our findings survive other robustness tests. In particular, they remain unchanged when we control for firm investment, financial soundness, uncertainty at the firm-, industry-, and country- level, GDP growth, election uncertainty, market expectations of future economic conditions, degree of social trust in the country, individual investors' tax rates, and strength of investor protection.

In a further quest to ascertain the sensitivity of our results to sample composition, we re-estimate our baseline model after excluding, sequentially, the U.S., Japan, and China, and considering different sample periods. We continue to document a positive and significant effect of EPU on dividends. Note that the effect remains positive and significant when we address endogeneity by implementing a two-stage instrumental variables analysis using political polarization to instrument EPU.⁶ We note further that dividends are firm-level variables, and are thus unlikely to impact the aggregate level of policy uncertainty in the economy ([Nagar et al. 2019](#)). Moreover, our findings survive a battery of robustness checks, including firm-fixed effects regression, accounting for a large set of additional control variables, and conducting a placebo test to further ease endogeneity concerns.

Additional analyses yield several other important findings. First, we find firms that are more vulnerable to agency problems (e.g., those with few growth opportunities and positive free cash flows) pay out more dividends when policy uncertainty is higher. Second (and delving deeper into the role of agency problems in the association between dividends and policy uncertainty), we find that EPU bears significantly positively on dividends in countries with weak shareholder protection, disclosure, securities regulation, and enforcement quality. According to prior research ([La Porta et al. 1998](#)), these countries have conditions that are more conducive to expropriation activities. In addition, the positive effect of EPU on dividends is less pronounced in countries with weak creditor rights, where managers need to restrict dividend payouts to accommodate creditors' concerns ([Brockman and Unlu 2009](#)).

Third, we consider agency conflicts embedded in ownership structures. In economies with weak shareholder protection, we document a stronger effect of EPU on dividends in family firms and in firms with concentrated ownership. These firms are likely to suffer higher agency problems, especially during bad times (e.g., [Bae et al. 2012](#); [Baek et al. 2004](#); [Boubakri et al. 2010](#); [Mitton 2002](#)). On a related note, firms with poor board governance characteristics (e.g., absence of an independent chairman, low director ownership, low director independence, and low external blockholdings) tend to pay higher dividends during periods of increased uncertainty over government policies. These new findings stress that dividend payout plays a relevant role in alleviating agency problems, lending further credence to the agency theory motive of corporate payout policy.

To summarize, in this paper, we report novel evidence of a positive association between dividends and policy-induced uncertainty. Central to our contribution are the new findings that agency problems and governance structures moderate the effect of policy-induced uncertainty on dividend policy. Our study also adds to the line of inquiry that stresses the importance of EPU in to the equity risk premium ([Brogaard and Detzel 2015](#)) as well as firm-level capital investment ([Gulen and Ion 2016](#)), investment efficiency ([Drobetz et al. 2018](#)), cash holdings ([Demir and Ersan 2017](#)), bank liquidity hoarding ([Berger et al. 2020](#)), merger and acquisition activity at both the firm and macro levels ([Bonaime et al. 2018](#)), and financial reporting quality ([El Ghouli et al. 2020](#)). Equally important, this study is just the first step of many needed to gain a deeper understanding of the factors that lead to different dividend payout policies over time and across countries.

⁵ This is one interpretation for the positive market reaction to an increase in dividend payout: dividends are an effective means to mitigate agency problems ([Easterbrook 1984](#); [Lippert et al. 2000](#)).

⁶ We are reluctant to claim causality between EPU and dividends because the magnitude of IV estimates is many times greater than that of the OLS estimates ([Jiang 2017](#)). In her meta-analysis study of 255 papers published in the top three finance journals, [Jiang \(2017, p. 131\)](#) cautions that the magnitude of IV estimates is predominantly greater than that of OLS estimates, and that "there is no reason to expect that the causal effects in close to 85% of all the cases studied by researchers should be predominantly higher than the simple correlational effect."

The rest of this study is organized as follows. In Section 2, we describe our research design and sample construction. Our empirical analysis and results are discussed in Sections 3 and 4. Section 5 concludes.

2. Data, variables, and descriptive statistics

2.1. Data and sample construction

We begin with all firms listed in Compustat North America and Compustat Global. We merge these files with the policy uncertainty index of 19 countries developed by BBD (2016) and macroeconomic variables from World Development Indicators (WDI). We exclude observations prior to 1991 due to limited country coverage. We further exclude financial firms (SIC codes 6000–6999) and utility firms (SIC codes 4900–4999) because of various regulatory restrictions that could affect their payout decisions. And we remove firm-year observations with negative values for total assets (AT), cash dividends declared on common/ordinary shares (DVC), sales (SALE), total common equity (CEQ), and those with missing values for SIC codes, cash dividends (DVC), and the control variables in our baseline model (see Section 2.2.3). We winsorize all firm-level ratio variables at the 1st and 99th percentiles to limit the influence of outliers, and we lag all control variables by one year to alleviate potential simultaneity concerns. Our initial sample is comprised of 190,799 firm-year observations for 23,042 unique firms from 19 countries during the 1991–2015 period.

2.2. Variables

2.2.1. Dividend policy

La Porta et al. (2000) consider the ratio of dividends to earnings to be the most commonly used measure of dividend payouts. Following Chay and Suh (2009), our main measure of a firm's dividend policy is the dividend payout ratio (*PAYOUT RATIO*), defined as the ratio of cash dividends declared on common/ordinary shares (DVC) to net income before extraordinary items (IB). We take *PAYOUT RATIO* to be missing if a dividend payer has negative earnings.

For robustness, we also consider several alternative measures of dividend policy when examining the relation between corporate dividends and policy uncertainty (e.g., Brockman and Unlu 2009; Floyd et al. 2015; Von Eije and Megginson 2008): dividend yield (*DIVIDEND YIELD*), dividends scaled by sales (*DIV_SALE*), dividends scaled by cash flow (*DIV_CF*), dividends per share (*DPS*), and total payout (*TOTAL PAYOUT RATIO*). *DIVIDEND YIELD* is defined as the ratio of cash dividends declared on common/ordinary shares (DVC) to firm market capitalization ($PRCC_F \times CSHO$). *DIV_SALE* and *DIV_CF* are the ratios of common stock cash dividends (DVC) to total sales (SALE) and to cash flow (i.e., net income (IB) plus depreciation and amortization (DP)), respectively. We treat *DIV_SALE* (*DIV_CF*) as missing if a dividend payer has negative sales (negative cash flow).

We compute *DPS* as cash dividends per share in \$U.S. to common stockholders, adjusted for securities identified as common stock equivalents and all stock splits/dividends that occurred during the fiscal period. The use of *DPS* helps reduce concerns about the choice of scale variable. Total payout (*TOTAL PAYOUT RATIO*) is the sum of cash dividends (DVC) and share repurchases divided by net income before extraordinary items (IB). *TOTAL PAYOUT RATIO* is treated as missing if a firm has negative earnings, unless the firm has zero dividends and stock repurchases. The repurchase amount equals purchases of common and preferred stock (PRSTKC) minus any decrease in the value of preferred stock outstanding (PSTKRV; if missing, then PSTK) (see Kahle and Stulz 2017).⁷ We set the repurchase amount to zero when it is missing to increase the sample size.

Finally, we examine investors' reactions to the announcement of dividend increases, *CAR_DIVINCR*, defined as the cumulative abnormal return over the period $[-1, +1]$ around the announcement of a dividend increase, where the cumulative abnormal return is the firm's stock return minus the country's value-weighted market return. We use CRSP for U.S. firms, and Compustat Global for non-U.S. firms.⁸

2.2.2. Economic policy uncertainty (EPU) index

Following recent literature (e.g., Bonaime et al. 2018; Brogaard and Detzel 2015; Gulen and Ion 2016), we capture economic policy-related uncertainty using the aggregate policy uncertainty index developed by BBD (2016). They construct a monthly EPU index for 19 countries⁹ by conducting a computer-automated search of major newspapers in each country to obtain a monthly count of articles containing the following terms: (E) "economic" or "economy"; (P) "tax," "government spending," "regulation," "central bank," or certain other policy-related terms; and (U) "uncertain" or "uncertainty." BBD then 1) scale the raw monthly EPU article counts by the total number of articles in the same newspaper and month, 2) standardize each newspaper's scaled EPU frequency count to a unit standard deviation, 3) average across the number of newspapers used in a country each month to obtain the EPU index for the country, and 4) normalize the resulting monthly series over the coverage period of the country to achieve a mean of 100. The final step yields the news-based EPU index for the 19 countries.

Note that the terms used in the news search for E, P, and U are carefully chosen by consulting professionals who are well versed in

⁷ Note that share repurchase is an alternative payout mechanism, which is more popular in the U.S. than in other countries in our sample. Specifically, when checking the repurchase amount by country, we find that the U.S. is the only country with a non-zero value at the 75th percentile, while 15 countries have a zero value at the 95th percentile. Thus, the regression results of *TOTAL PAYOUT RATIO* must be interpreted with caution.

⁸ We thank anonymous reviewers for suggesting these alternative measures of dividends.

⁹ The index has been made available by the authors at <http://www.policyuncertainty.com/index.html>.

the local language and economy, and who can identify the uncertainty specific to each country. BBD show that their index captures clear spikes in uncertainty around important policy-relevant events, such as the Gulf Wars and the debt-ceiling dispute in the summer of 2011. Their index is not necessarily correlated with political events that have milder economic consequences.

To rule out concerns that the newspaper-based index could be an inaccurate measure of the overall level of policy-induced uncertainty, BBD conduct various validation tests. First, under close supervision and training, they employ human audits of newspapers. They find that their computer-automated search is strongly correlated with the results of a human-generated index. Second, to ensure that a newspaper's political slant does not significantly affect reliability, they use the media slant index of Gentzkow and Shapiro (2010) to categorize newspapers as exhibiting a left or right political slant. They then compare the "left" and "right" versions of the index, and find that, regardless of political slant, their index does not distort variations in policy uncertainty over time.

Third, BBD compare their index to other reasonable measures of economic uncertainty, such as the Chicago Board Options Exchange Volatility Index, and to indicators based on an analysis of the Beige Book and 10-K filings. They confirm their index is distinct in scope from other indicators, and that it contains information about policy-related economic uncertainty as opposed to general financial uncertainty and stock market events. We also note that premier commercial data providers such as Bloomberg, Haver Analytics, and Reuters carry the BBD index, suggesting it is relevant to the entities (e.g., banks, hedge funds, and policymakers) that subscribe to these data services.

Following Gulen and Ion (2016), we compute economic policy uncertainty (*EPU*) as the natural logarithm of the 12-month arithmetic average of the BBD index ending in the fiscal year-end month.

2.2.3. Control variables

Closely following the literature on dividend policy (e.g., Attig et al. 2016; Brockman and Unlu 2009), we include six variables to control for firm-level characteristics: 1) cash flow uncertainty (*SRVOL*), the standard deviation of monthly stock returns in the current fiscal year; 2) retained earnings (*RE*), the ratio of retained earnings (*RE*) to common stockholders' equity (*CEQ*); 3) equity ratio (*TE*), the ratio of common stockholders' equity (*CEQ*) to total assets (*AT*); 4) profitability (*ROA*), the ratio of net income before extraordinary items (*IB*) to total assets (*AT*); 5) natural logarithm of sales growth (*SGR*), $\log[(\text{SALE}_t/\text{SALE}_{t-1})]$; 6) firm size (*SIZE*), the natural logarithm of total assets (*AT*) in millions of \$U.S.; and 7) cash holdings (*CASH*), the ratio of cash and short-term investments (*CHE*) to total assets (*AT*). In addition, we include GDP per capita (*GDP*), defined as the natural logarithm of GDP per capita in constant 2010 \$U.S. from WDI to control for the aggregate economic environment.

Table 1 summarizes definitions and data sources for all variables used in our analyses.

2.3. Descriptive statistics

Table 2 presents summary statistics for our initial sample of 190,799 firm-year observations. To reduce concerns about potential sample bias when examining the relation between dividend payouts and policy uncertainty, we do not restrict our sample according to the availability of our primary payout measure (*PAYOUT RATIO*). When we employ *PAYOUT RATIO*, our main sample comprises 179,186 firm-year observations for 22,802 firms from 19 countries over the 1991–2015 period. The mean (median) *PAYOUT RATIO* is 0.28 (0.13), with a range from 0 at the 5th percentile to 1 at the 95th percentile.

Table 3 presents descriptive statistics for our main sample by country. On average, we observe considerable variation in *PAYOUT RATIO* across countries. Over our sample period, Australia exhibits the highest average *PAYOUT RATIO* of 0.72, followed by the U.K. (0.59), and Netherlands and Brazil (0.57), while China and the U.S. exhibit the lowest, at 0.16 and 0.13, respectively. The average level of policy uncertainty (*EPU*), the natural logarithm of the original BBD index, is highest for Brazil, at 4.91, and lowest for Ireland, at 4.45. In line with other cross-country studies, U.S. and Japanese firms account for 45% and 22% of our main sample, respectively.¹⁰ In the next section, we show that our findings are not driven by the dominance of these countries in our sample.

Table 4 shows the distribution of our main sample by both year and industry, where we use the Fama–French 12-industry classification. Our sample reveals decent variations in *PAYOUT RATIO* across industries, from 0.37 in consumer non-durables to 0.14 in the healthcare, medical equipment, and drug industries.

Turning to the evolution of *PAYOUT RATIO* over time, we find that, beginning with 0.26 in 1991, *PAYOUT RATIO* fell to a low of 0.21 in 1995, rose to 0.27 in 1998 (the Asian financial crisis), and peaked at 0.30 in 2001 (9/11 and the bursting of the dot-com bubble), and then fluctuated at around 0.28 until 2007. It then jumped to 0.34 in 2008 (global financial crisis), sank to 0.27 in 2011, rebounded to 0.29 in 2012 (Eurozone crises, U.S. fiscal fights, China leadership transition, European immigration crisis), and continued to rise to around 0.37 in 2014 and 2015 (continuing Eurozone crisis, China growth slowdown concerns).

Fig. 1 illustrates the comovements between *PAYOUT RATIO* and economic policy uncertainty over the 1991–2015 period. The correlation between *PAYOUT RATIO* and major events contributing to global *EPU* provides preliminary support for the prediction of higher dividend payouts during high-*EPU* periods.

¹⁰ Brockman and Unlu (2009) point out that dominance of U.S., U.K., and Japanese firms is common in international studies. In our sample, U.K. firms account for only 4.4% of observations, in part because the monthly BBD index for the U.K. is only available from 1998.

Table 1
Variable definitions.

Variable	Definition	Source
<i>Dependent Variables</i>		
<i>PAYOUT RATIO</i>	Ratio of cash dividends declared on common/ordinary shares (DVC) to net income before extraordinary items (IB).	Authors' calculations based on Compustat data
<i>DIVIDEND YIELD</i>	Ratio of cash dividends declared on common/ordinary shares (DVC) to firm market capitalization ($PRCC_F \times CSHO$).	As above
<i>DIV_SALE</i>	Ratio of cash dividends declared on common/ordinary shares (DVC) to total sales (SALE).	As above
<i>DIV_CF</i>	Ratio of cash dividends declared on common/ordinary shares (DVC) to the sum of net income (IB) plus depreciation and amortization (DP).	As above
<i>DPS</i>	Cash dividend per share in \$U.S. to common stockholders, defined as cash dividends declared on common/ordinary shares (DVC) divided by average or actual number of common/ordinary shares outstanding, adjusted for securities identified as common/ordinary stock equivalents as reported by the company (CSHPRIA/CSHPRI), adjusted for all stock splits and stock dividends that occurred during the period (AJEXI/AJEX).	As above
<i>TOTAL PAYOUT RATIO</i>	Ratio of cash dividends declared on common/ordinary shares (DVC) and repurchases to net income before extraordinary items (IB). Repurchase amount equals purchase of common and preferred stock (PRSTKC) minus any decrease in the value of preferred stock outstanding (PSTKR; if missing, then PSTK).	As above
<i>CAR_DIVINCR</i>	Cumulative abnormal return over the period $[-1, +1]$ around the announcement of a dividend increase. The cumulative abnormal return is the firm's stock return minus the country's value-weighted market return.	CRSP for U.S. firms and Compustat Global for non-U.S. firms.
<i>DPS Initiation</i>	A dummy that equals 1 if the firm has positive <i>DPS</i> in year <i>t</i> but 0 <i>DPS</i> in year <i>t</i> -1, and 0 otherwise.	Authors' calculations based on Compustat data
<i>DPS Increase</i>	A dummy variable that equals 1 if the firm has increased <i>DPS</i> in year <i>t</i> relative to year <i>t</i> -1, and 0 otherwise.	Authors' calculations based on Compustat data
<i>Key Explanatory Variable</i>		
<i>EPU</i>	Natural logarithm of the moving average of the monthly policy uncertainty index over the 12 months ending in the month of the fiscal year-end.	BBD (2016)
<i>Control Variables</i>		
<i>SRVOL</i>	Standard deviation of monthly stock returns in the current fiscal year.	Authors' calculations based on Compustat data
<i>RE</i>	Ratio of retained earnings (RE) to common stockholders' equity (CEQ).	As above
<i>TE</i>	Ratio of common stockholders' equity (CEQ) to total assets (AT).	As above
<i>ROA</i>	Return on assets, computed as ratio of net income before extraordinary items (IB) to total assets (AT).	As above
<i>SGR</i>	Natural logarithm of sales growth, computed as $\log[(SALE_t/SALE_{t-1})]$ or $\log(\frac{\Delta SALE_t}{SALE_{t-1}} + 1)$.	As above
<i>SIZE</i>	Natural logarithm of total assets (AT) in millions of \$U.S.	As above
<i>CASH</i>	Ratio of cash and short-term investments (CHE) to total assets (AT).	As above
<i>GDPPC</i>	Natural logarithm of GDP per capita in constant 2010 \$U.S.	WDI
<i>Variables for Additional Analyses</i>		
<i>CAPX</i>	Ratio of capital expenditures (CAPX) to total assets (AT).	Authors' calculations based on Compustat data
<i>XRD</i>	Ratio of R&D expenditures (XRD) to total assets (AT).	As above
<i>XRDDUM</i>	Binary variable that equals 1 if R&D expenditure (XRD) is non-zero or non-missing, and 0 otherwise.	As above
<i>FCF</i>	Sum of net income (IB) plus depreciation and amortization (DP) divided by total assets (AT).	As above
<i>MTB</i>	Ratio of market value of assets to total book assets, where market value of assets equals total book assets minus stockholder's equity plus market capitalization.	As above
<i>LEVERAGE</i>	Ratio of total long-term liabilities (DLTT) to total assets (AT).	As above
<i>IO</i>	Percentage of shareholding by institutional investors.	FactSet LionShares
<i>EARNVOL</i>	Five-year standard deviation of firm's annual earnings (IB) from year <i>t</i> -4 to <i>t</i> .	As above
<i>INDUSTRY_SHOCK</i>	First principal component of the industry-year median of the absolute change in each of the seven economic shock variables, including profitability, asset turnover, R&D, capital expenditures, employee growth, ROA, and sales growth, using the entire Compustat universe.	As above
<i>SD_SALES_GR</i>	Cross-sectional standard deviation of sales growth, calculated for each country-year, using the entire Compustat universe.	As above
<i>GDPGROWTH</i>	Annual percentage of GDP growth.	WDI
<i>ELECTION</i>	Binary variable that equals 1 if there was an executive election or a legislative election in a given year, and 0 otherwise.	Database of Political Institutions
<i>R_GDP_F</i>	Real GDP growth rate forecast based on an assessment of the economic climate in individual countries and the world economy using a combination of model-based analyses and expert judgment. This indicator is measured as a year-over-year growth rate.	OECD
<i>CLI</i>	Moving average of the monthly composite leading indicator of turning points in business cycles over the 12 months ending in the month of the fiscal year-end. The index shows a fluctuation of economic activity around its long-term potential level. It indicates short-term economic movements in qualitative rather than quantitative terms.	As above

(continued on next page)

Table 1 (continued)

Variable	Definition	Source
CCI	Moving average of the monthly consumer confidence index over the 12 months ending in the month of the fiscal year-end. The index is based on households' plans for major purchases and their economic situation, both currently and in the immediate future. Opinions compared to a "normal" state are collected, with the difference between positive and negative answers providing a qualitative index on economic conditions.	As above
TRUST	Percentage of survey responses indicating that most people can be trusted in reply to the WVS question, "Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?" This variable is constructed as an average value by country and by year, using information from five waves of surveys over the 1989–2014 period. For gap years when the survey was not conducted for a given country, the value from the most recent survey is used.	World Values Survey
DIVTAX	Maximum statutory dividend tax rates for individuals (in percent).	Hail et al. (2017)
CGTAX	Maximum statutory capital gains tax rates for individuals (in percent).	As above
SHAREHOLDER PROTECT	First principal component of revised anti-director rights index and law and order index. The revised anti-director rights index measures the extent to which minority shareholders are protected in the corporate decision process, in terms of the right to vote. It covers the following six areas: "(1) vote by mail; (2) obstacles to the actual exercise of the right to vote (i.e., the requirement that shares be deposited before the shareholders' meeting); (3) minority representation on the board of directors through cumulative voting or proportional representation; (4) an oppressed minority mechanism to seek redress in case of expropriation; (5) preemptive rights to subscribe to new securities issued by the company; and (6) the right to call a special shareholder meeting." Law and order index measures the strength and impartiality of the legal system and assesses the popular observance of the law.	Djankov et al. (2008b) and ICRG
CREDITOR PROTECT	First principal component of the creditor rights index and the recovery rate. The index of creditor rights measures four powers of secured lenders in bankruptcy granted by a country's laws and regulations. The index ranges from 0 (weak) to 4 (strong). This variable is available for the 1990–2003 period. Values in subsequent years are set to the values of the 2003 variable. The recovery rate for secured creditors is defined as the present value of the terminal value of the firm after bankruptcy costs, considering any violations of the absolute priority rule. The recovery rate is a percentage ranging from 0 to 1.	Djankov et al. (2007) and Djankov et al. (2008a)
POLAR	Product of probability, that two deputies picked at random from the legislature will be of different parties, and government opposition constructed as 1 minus government support (or the government share of seats in parliament).	Database of Political Institutions
POSFCF	Binary variable that equals 1 if the firm has positive free cash flows, and 0 otherwise. Free cash flow is defined as net income (IB) plus depreciation and amortization (DP) scaled by total assets (AT).	Authors' calculations based on Compustat data
DISCLOSE	Arithmetic mean of six indicators of prospectus disclosure requirements (i.e., prospects, compensation, shareholders, insider ownership, irregular contracts outside the ordinary course of business, and transactions with related parties).	La Porta et al. (2006)
SECURITY_REG	First principal component of the disclosure index, the liability standard index, and the public enforcement index, which measure disclosure requirements and liability standards that facilitate private enforcement, as well as the strength of public enforcement by supervisory bodies.	As above
PUBLIC ENFORCE	Index of public enforcement including fines and prison terms for related parties in self-dealing if all disclosure and approval requirements have been met, varying from 0 to 1.	Djankov et al. (2008b)
Blockholder	Dummy that equals 1 if the firm has a blockholder holding more than 25% of the shares and 0 otherwise.	Osiris (Bureau van Dijk)
Family	Dummy that equals 1 if blockholder type is "individual(s)/family(ies)," and 0 otherwise.	As above
INDEP_CHAIRMAN	Binary variable that equals 1 if the board chair is independent, and 0 otherwise.	Authors' calculations based on NRG
BOARD_OWNERSHIP	Percentage of shares held by board members.	NRG
BOARD_INDEP	Ratio of independent members to board size.	As above
EXTERNALBLOCK	Percentage of shares held by external blockholders.	As above
BOARD_INDEP_EXPAND	Ratio of independent members to board size.	ISS, BOARDEX, NRG, and ASSET4

3. Empirical analysis

3.1. Policy uncertainty and dividend payouts

Our baseline model is as follows:

$$\text{Payout variable}_{i,j,t} = \beta_0 \text{EPU}_{j,t} + \beta_1 \text{SRVOL}_{i,t-1} + \beta_2 \text{RE}_{i,t-1} + \beta_3 \text{TE}_{i,t-1} + \beta_4 \text{ROA}_{i,t-1} + \beta_5 \text{SGR}_{i,t-1} + \beta_6 \text{SIZE}_{i,t-1} + \beta_7 \text{CASH}_{i,t-1} + \beta_8 \text{GDPPC}_{j,t-1} + \delta_i + \text{FIXED EFFECTS} + \varepsilon_{i,t},$$

where i , j , and t index firms, industries, and years, respectively. The dependent variable is one of the payout variables described in Section 2.2.1. *FIXED EFFECTS* contains industry, country, or firm fixed effects depending on the regression. Our cross-country setting enables us to include year fixed effects, δ_t , in all regressions; this is not feasible in single-country studies of policy uncertainty, where all

Table 2
Summary statistics for the initial sample.

Variables	N	Mean	SD	p5	p25	p50	p75	p95
PAYOUT_RATIO	179,186	0.283	0.514	0.000	0.000	0.128	0.360	1.002
DIVIDEND_YIELD	189,061	0.015	0.019	0.000	0.000	0.008	0.022	0.053
DIV_SALE	190,648	0.013	0.024	0.000	0.000	0.004	0.014	0.057
DIV_CF	179,832	0.130	0.190	0.000	0.000	0.066	0.189	0.494
DPS	180,717	0.180	0.420	0.000	0.000	0.024	0.156	0.880
CAR_DIVINCR	56,459	0.009	0.050	-0.071	-0.019	0.005	0.033	0.105
TOTAL_PAYOUT_RATIO	173,627	0.410	0.765	0.000	0.000	0.196	0.468	1.528
Raw EPU	190,799	110.418	36.466	67.136	82.921	105.432	128.339	170.791
EPU	190,799	4.655	0.311	4.207	4.418	4.658	4.855	5.140
SRVOL	190,799	0.129	0.079	0.044	0.076	0.109	0.158	0.283
RE	190,799	-0.065	1.896	-2.617	0.035	0.364	0.659	0.956
TE	190,799	0.508	0.211	0.168	0.349	0.497	0.670	0.864
ROA	190,799	0.011	0.139	-0.238	0.006	0.032	0.068	0.146
SGR	190,799	0.107	0.399	-0.266	-0.016	0.072	0.193	0.582
SIZE	190,799	5.662	1.916	2.575	4.414	5.572	6.830	9.030
CASH	190,799	0.178	0.181	0.007	0.047	0.120	0.244	0.579
GDPPC	190,799	10.237	1.005	7.553	10.507	10.649	10.724	10.819
CAPX	167,757	0.056	0.059	0.004	0.017	0.037	0.072	0.178
XRD	179,186	0.028	0.066	0.000	0.000	0.000	0.021	0.160
XRDDUM	179,186	0.441	0.496	0.000	0.000	0.000	1.000	1.000
FCF	172,359	0.051	0.143	-0.211	0.035	0.071	0.114	0.199
MTB	177,533	1.785	1.373	0.712	0.996	1.323	2.002	4.543
LEVERAGE	178,946	0.121	0.141	0.000	0.001	0.069	0.196	0.416
IO	87,929	0.259	0.297	0.001	0.031	0.117	0.421	0.892
EARNVOL	173,161	42.768	125.973	0.558	2.302	6.746	22.924	195.016
INDUSTRY_SHOCK	179,175	0.532	1.578	-1.308	-0.457	0.326	1.117	3.990
SD_SALES_GR	179,040	16.561	32.434	0.125	0.270	1.216	20.726	53.293
GDPGROWTH	179,186	3.295	3.216	-0.956	1.608	2.746	4.280	9.801
ELECTION	176,101	0.349	0.477	0.000	0.000	0.000	1.000	1.000
R_GDP_F	176,986	3.229	3.121	-0.627	1.602	2.742	4.192	9.651
CLI	176,984	99.972	1.074	98.164	99.367	100.120	100.690	101.434
CCI	168,282	99.903	1.381	97.345	98.906	99.964	100.817	102.235
TRUST	157,369	0.393	0.092	0.233	0.359	0.388	0.423	0.545
DIVTAX	179,186	24.917	12.917	0.000	15.000	25.000	39.600	39.900
CGTAX	179,186	18.642	9.745	0.000	15.000	20.000	26.000	37.700
SHAREHOLDER_PROTECT	179,186	-0.032	1.093	-2.684	-0.537	0.402	0.440	1.379
CREDITOR_PROTECT	170,972	-0.043	1.014	-1.479	-0.453	-0.453	0.749	1.546
POLAR	175,911	0.226	0.101	0.000	0.191	0.248	0.275	0.347
POSFCF	172,359	0.862	0.344	0.000	1.000	1.000	1.000	1.000
DISCLOSE	160,160	0.880	0.147	0.667	0.750	1.000	1.000	1.000
SECURITY_REG	160,160	0.055	1.601	-2.096	-1.701	1.456	1.456	1.456
PUBLIC_ENFORCE	179,186	0.118	0.279	0.000	0.000	0.000	0.000	1.000
INDEP_CHAIRMAN	10,796	0.336	0.472	0.000	0.000	0.000	1.000	1.000
BOARD_OWNERSHIP	11,206	0.057	0.139	0.000	0.000	0.007	0.036	0.315
BOARD_INDEP	11,206	0.640	0.274	0.000	0.500	0.710	0.870	0.920
EXTERNALBLOCK	11,206	0.073	0.178	0.000	0.000	0.000	0.032	0.520
BOARD_INDEP_EXPAND	49,918	0.628	0.252	0.000	0.500	0.667	0.833	1.000

This table reports the number of observations, means, standard deviations, and the 5th, 25th, 50th, 75th, and 95th percentiles for the variables used in this paper. The initial sample consists of 190,799 firm-year observations for 23,042 unique firms from 19 countries during the 1991–2015 period. Variable definitions and sources are in [Table 1](#).

firms face the same level of country-specific policy uncertainty in a given period (e.g., [Nguyen and Phan 2017](#)). Throughout this paper, standard errors are clustered at the firm level to adjust for within-firm correlation.¹¹ β_0 can be interpreted as changes in dependent variables when policy uncertainty (raw EPU before logarithmic transformation) doubles (increases by 100%) while holding other variables at their mean values.

[Table 5](#) presents our main evidence on the relation between policy uncertainty and corporate dividend payouts. In Models 1 and 2, we focus on PAYOUT_RATIO. We estimate the impact of EPU on dividends after controlling for firm-level characteristics and macroeconomic conditions. In Model 1, we control for firm and year fixed effects. This model removes any purely cross-sectional

¹¹ [Cameron and Miller \(2015\)](#) suggest that having less than 20 clusters in a balanced panel data can cause the problem of “too few clusters” and lead to biased standard errors, which is more problematic in the unbalanced case. Because we only have 19 countries, and even fewer in the robustness and subsample tests, clustering standard errors at the country level can bias standard errors ([Cameron and Miller 2015](#); [Petersen 2009](#); [El Ghoul et al. 2020](#)). Nevertheless, for robustness, we cluster standard errors at the country level and use two-way clustering at both the country-year and firm level. The results are in [Table 9](#).

Table 3
Summary statistics by country.

Country	N	%	Unique firms	PAYOUT RATIO	EPU	SRVOL	RE	TE	ROA	SGR	SIZE	CASH	GDPPC
Australia	3467	1.9%	610	0.72	4.48	0.10	0.09	0.52	0.07	0.14	5.44	0.12	10.80
Brazil	731	0.4%	162	0.57	4.91	0.12	0.29	0.48	0.07	0.16	6.93	0.17	9.30
Canada	2985	1.7%	440	0.17	4.69	0.15	−0.81	0.56	−0.04	0.14	5.77	0.17	10.67
Chile	104	0.1%	44	0.42	4.46	0.08	0.44	0.54	0.06	0.13	6.41	0.08	9.30
China	18,924	10.6%	2424	0.16	4.81	0.13	0.05	0.53	0.04	0.15	5.80	0.21	8.23
France	1279	0.7%	363	0.52	4.89	0.09	0.20	0.43	0.05	0.08	6.38	0.16	10.60
Germany	3338	1.9%	512	0.50	4.70	0.09	0.31	0.40	0.04	0.07	6.34	0.13	10.58
India	8214	4.6%	1521	0.28	4.66	0.14	0.63	0.48	0.08	0.16	4.85	0.11	7.16
Ireland	458	0.3%	51	0.38	4.45	0.09	0.20	0.44	0.05	0.12	5.98	0.15	10.53
Italy	819	0.5%	187	0.48	4.63	0.09	0.23	0.43	0.03	0.09	6.54	0.13	10.49
Japan	40,102	22.4%	3497	0.49	4.62	0.10	0.53	0.47	0.03	0.04	6.29	0.18	10.67
Netherlands	426	0.2%	95	0.57	4.49	0.08	0.32	0.41	0.06	0.06	7.00	0.10	10.81
Russia	102	0.1%	42	0.37	4.63	0.13	0.58	0.54	0.08	0.15	7.61	0.07	9.13
Singapore	2197	1.2%	452	0.51	4.66	0.11	0.45	0.57	0.07	0.10	5.20	0.21	10.71
South Korea	5030	2.8%	853	0.32	4.73	0.13	0.12	0.49	0.05	0.10	6.20	0.15	9.90
Spain	431	0.2%	89	0.45	4.59	0.08	0.34	0.40	0.06	0.08	7.10	0.10	10.33
Sweden	2554	1.4%	355	0.52	4.49	0.11	0.28	0.47	0.03	0.10	5.48	0.15	10.77
United Kingdom	7835	4.4%	1342	0.59	4.65	0.10	0.25	0.47	0.06	0.10	5.65	0.12	10.52
United States	80,190	44.8%	9763	0.13	4.63	0.15	−0.61	0.54	−0.02	0.14	5.24	0.20	10.68
Total	179,186	100.0%	22,802	0.28	4.65	0.13	−0.09	0.51	0.01	0.11	5.63	0.18	10.22

This table reports the number of observations and firms, as well as the averages of the regression variables, by country. The final sample for our main regression includes 179,186 firm-year observations for 22,802 unique firms from 19 countries over the 1991–2015 period. Variable definitions and sources are in [Table 1](#).

Table 4
Statistics by year and industry.

Panel A. Sample distribution by year				EPU	Panel B. Industry distribution			
Year	N	%	PAYOUT RATIO		Campbell	N	%	PAYOUT RATIO
1991	4103	2.29%	0.26	4.69	BusEq	31,811	17.75%	0.18
1992	4182	2.33%	0.27	4.72	Chems	9303	5.19%	0.35
1993	4431	2.47%	0.26	4.63	Durbl	8515	4.75%	0.30
1994	4958	2.77%	0.22	4.49	Enrgy	5849	3.26%	0.23
1995	5297	2.96%	0.21	4.45	Hlth	15,986	8.92%	0.14
1996	5961	3.33%	0.22	4.37	Manuf	33,118	18.48%	0.34
1997	6270	3.50%	0.22	4.48	NonDur	16,372	9.14%	0.37
1998	6723	3.75%	0.27	4.70	Other	30,664	17.11%	0.32
1999	6765	3.78%	0.25	4.39	Shops	23,804	13.28%	0.31
2000	6755	3.77%	0.26	4.49	Telcm	3764	2.10%	0.33
2001	6608	3.69%	0.30	4.85				
2002	7421	4.14%	0.29	4.79				
2003	7825	4.37%	0.27	4.74				
2004	8450	4.72%	0.27	4.40	Total	179,186	100.00%	0.28
2005	8262	4.61%	0.30	4.24				
2006	8419	4.70%	0.29	4.22				
2007	8418	4.70%	0.28	4.38				
2008	7852	4.38%	0.34	4.93				
2009	7874	4.39%	0.31	4.82				
2010	8406	4.69%	0.28	4.85				
2011	8641	4.82%	0.27	5.07				
2012	8957	5.00%	0.29	5.14				
2013	9398	5.24%	0.27	4.81				
2014	8777	4.90%	0.37	4.63				
2015	8433	4.71%	0.37	4.78				
Total	179,186	100.00%	0.28	4.65				

This table reports the distribution of firms over time and across industries for our main sample of 179,186 firm-year observations for 22,802 unique firms from 19 countries over the 1991–2015 period. Panel A reports changes in the mean value of *PAYOUT RATIO* and *EPU* by year. Panel B reports *PAYOUT RATIO* by industry. *PAYOUT RATIO* is defined as the ratio of cash dividends to common stockholders (DVC) to net income before extraordinary items (IB).

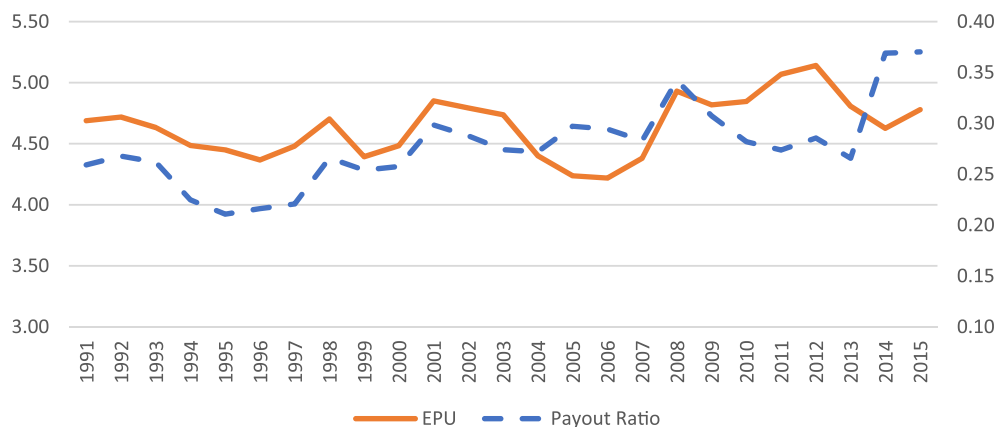


Fig. 1. Changes in dividend payout and EPU over time.

correlation between *EPU* and *PAYOUT RATIO* with firm-fixed effects and provides time series evidence for the impact of *changes* in policy uncertainty on *changes* corporate dividend policy.

Khan et al. (2016) argue that the inclusion of both firm and time fixed effects is a generalization of the difference-in-differences approach, which improves causal interpretation. Combined with standard errors clustered at the firm level, this model accounts for correlations among unobservable time-invariant firm effects and explanatory variables, unobservable aggregate time trends, and within-firm serial correlation.

We find that the estimated coefficient on *EPU* is positive (0.103) and statistically significant at the 1% level. Consistent with the agency perspective, paying dividends during high-*EPU* periods reduces the agency costs of free cash flow, because *EPU* tends to depress investment opportunities and exacerbate agency problems. Interestingly, if we hold other variables at their sample mean, a one-standard-deviation increase in raw *EPU* (36.5) above its mean (110.4), equivalent to its increase by 33% (36.5/110.4), corresponds

Table 5
Policy uncertainty and dividend payout.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	<i>PAYOUT_RATIO</i>		<i>DIVIDEND_YIELD</i>	<i>DIV_SALE</i>	<i>DIV_CF</i>	<i>DPS</i>	<i>TOTAL_PAYOUT_RATIO</i>	<i>DPS (for dividend payer only)</i>	<i>CAR_DIVINCR</i>
	Firm Fixed Effects	Country Fixed Effects	Alternative payout measures						
<i>EPU</i>	0.103*** (12.165)	0.111*** (12.927)	0.003*** (9.282)	0.003*** (8.624)	0.033*** (12.709)	0.012** (2.359)	0.069*** (5.823)	0.016** (2.205)	0.002** (2.007)
<i>SRVOL</i>	−0.331*** (−19.126)	−0.648*** (−33.731)	−0.016*** (−28.417)	−0.011*** (−15.015)	−0.126*** (−20.457)	−0.098*** (−8.018)	−0.703*** (−23.261)	−0.175*** (−9.270)	−0.008 (−1.116)
<i>RE</i>	−0.007*** (−8.309)	0.006*** (8.920)	−0.000*** (−3.301)	−0.001*** (−10.772)	−0.003*** (−7.438)	−0.001 (−0.840)	−0.017*** (−10.028)	0.026*** (4.169)	−0.001 (−0.838)
<i>TE</i>	0.201*** (14.004)	0.031*** (2.978)	0.005*** (10.626)	0.012*** (16.128)	0.095*** (17.538)	0.089*** (6.849)	0.485*** (19.910)	0.083*** (3.188)	−0.012*** (−3.717)
<i>ROA</i>	−0.090*** (−9.211)	0.012 (1.307)	0.002*** (6.549)	0.009*** (15.426)	0.022*** (5.496)	0.066*** (7.688)	−0.005 (−0.266)	0.623*** (16.113)	−0.010 (−1.322)
<i>SGR</i>	−0.029*** (−11.928)	−0.056*** (−21.888)	−0.000*** (−3.518)	−0.001*** (−7.678)	−0.008*** (−10.334)	0.004** (2.481)	−0.068*** (−14.629)	−0.004 (−0.830)	0.000 (0.307)
<i>SIZE</i>	0.063*** (18.756)	0.020*** (15.418)	0.001*** (7.125)	0.001*** (8.377)	0.014*** (11.228)	0.046*** (13.694)	0.162*** (27.442)	0.128*** (19.167)	−0.007*** (−8.785)
<i>CASH</i>	0.028** (1.999)	0.016 (1.544)	−0.000 (−0.572)	0.007*** (9.346)	0.038*** (6.731)	0.086*** (8.278)	0.138*** (5.362)	0.143*** (6.984)	0.007* (1.815)
<i>GDP_PC</i>	−0.236*** (−17.089)	−0.146*** (−11.198)	−0.016*** (−29.969)	−0.019*** (−19.761)	−0.140*** (−23.672)	−0.186*** (−19.757)	−0.315*** (−17.580)	−0.299*** (−15.864)	−0.001 (−0.323)
Observations	179,186	179,186	189,061	190,648	179,832	180,717	173,627	105,088	56,459
Firm FE	YES	NO	YES	YES	YES	YES	YES	YES	YES
CN FE	NO	YES	NO	NO	NO	NO	NO	NO	NO
Industry FE	NO	YES	NO	NO	NO	NO	NO	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj R2	0.304	0.159	0.663	0.690	0.550	0.744	0.195	0.816	0.082

This table reports results from regressing different measures of dividend policy on *EPU*, as well as firm- and country-level controls using the initial sample of 190,799 firm-year observations for 23,042 unique firms from 19 countries over the 1991–2015 period. The sample size varies across models due to the availability of the dependent variable used. In Models 1 and 2, the dependent variable is *PAYOUT_RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). The dependent variables are *DIVIDEND_YIELD*, the ratio of cash dividends (DVC) to firm market capitalization, in Model 3; *DIV_SALE*, the ratio of cash dividends (DVC) to total sales (SALE), in Model 4; *DIV_CF*, the ratio of cash dividends (DVC) to the sum of net income (IB) plus depreciation and amortization (DP), in Model 5; *DPS*, cash dividend per share in U.S. to common stockholders, in Models 6 and 8; *TOTAL_PAYOUT_RATIO*, the sum of cash dividends (DVC) and repurchases divided by net income before extraordinary items (IB), in Model 7; and *CAR_DIVINCR*, the cumulative abnormal return [−1, +1] around the announcement of dividend increase, in Model 9. Model 8 uses a sample including only firm-year observations with positive *DPS*. All variables are defined in Table 1. Year dummies are included in all regressions. Industry dummies based on the Fama–French 12-industry classification, country fixed effects, and firm fixed effects are included as indicated in Table 5. All parameters are estimated using pooled OLS, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

to a 0.034 (0.33×0.103) increase in *PAYOUT RATIO*. Given that the unconditional mean *PAYOUT RATIO* is 0.283, the magnitude of this effect (12%) is economically sizable.¹²

In Model 2, we replace firm fixed effects with industry and country fixed effects. Country fixed effects control for unobservable, time-invariant, country-level heterogeneity that may correlate with the level of policy uncertainty or influence corporate payout practices in a given country. Consistent with our previous finding, our main variable of interest, *EPU*, enters the regression positively, and is significant at the 1% level. This result suggests that firms tend to increase dividend payouts as policy uncertainty increases.

In Models 3–7, we rerun the specification in Model 1 using the alternative payout measures (*DIVIDEND_YIELD*, *DIV_SALE*, *DIV_CF*, *DPS*, and *TOTAL PAYOUT RATIO*, respectively). We find that the coefficients on *EPU* are positive and significant at the 5% level or better in all five models. The results indicate that, when facing a high level of policy-related uncertainty, firms tend to have higher dividend-to-price ratios, dividend-to-sales ratios, dividend-to-cash flow ratios, dividends per share, and total payout ratios (including both dividends and repurchases).¹³ These results are consistently in line with our previous finding based on *PAYOUT RATIO*, alleviating concerns that the results are picking up the relation between the scaling variables and *EPU*.

Note that an important clarification about this evidence is in order. *EPU* has greater explanatory power for the payout ratio measured by regular dividends (Models 1 and 2) than for total payout, which includes repurchases (Model 7). At first glance, this result seems surprising because share repurchases tend to preserve a firm's financial flexibility (see Farre-Mensa et al. 2014, and the references therein). This is because repurchases do not compel the firm to commit to a future payout (Farre-Mensa et al. 2014, and the references therein) and their per share amounts are not typically reported (Baker et al. 2015), making repurchases difficult to be assessed by investors. In addition, unlike dividends, which tend to be associated with relatively permanent operating cash flow shocks (Jagannathan et al. 2000), repurchases are used to distribute more transient earnings (Farre-Mensa et al. 2014). Thus, on a prima facie basis, one would expect *EPU* to have a more pronounced impact on total payout given the potential stronger effect on repurchases.

However, several observations appear to lend credence to our evidence that the effect of *EPU* is more pronounced on dividends than on total payout. First, while in a frictionless market, dividends and repurchases may be substitutes, they are not perfect substitutes in practice (e.g., Jagannathan et al. 2000, among others). During high-*EPU* periods (and their associated increased frictions, as discussed at the outset), dividends appear to be a more credible monitoring device in addressing agency problems and limiting management opportunism. This is because, unlike repurchases, dividends are clearly reported, can be easily tracked by investors (or other market participants), serve as reference points for investors (Baker and Wurgler, 2015), and thus are less subject to managerial discretion.

Second, the evidence that *EPU* has a larger effect on dividends than on total payout is also – to a large extent – compatible with the argument that an increase in uncertainty may increase investors' preference for cash dividends as an effective means of mitigating agency problems (Easterbrook 1984; Lippert et al. 2000; Faccio et al. 2001). Third, since repurchases tend to be procyclical (e.g., Jagannathan et al. 2000), repurchase programs are expected to decline during high-*EPU* periods. Fourth, anecdotal evidence points to concerns over debt-financing of repurchases (Shorter 2019), which may increase the financial risk of firms, and thus is less desirable to investors, all else equal.

Fifth, repurchases can be exploited by managers for personal financial gain. For instance, managers may favor repurchases over dividends to meet earnings per share (EPS) targets, since their compensation packages tend to be tied to EPS. Bonaimé et al. (2016) suggest that the cost of financial flexibility associated with share repurchases is greater when managers use repurchases to beat analysts' EPS forecasts and when corporate governance is weak. Further, as suggested by Fried (2005), repurchases can be associated with some forms of managerial opportunism (e.g., informed trading, false signaling, expropriation of investors). For instance, Brockman et al. (2008) find that managers alter information flows around share repurchases and that managers' propensity to use opportunistic disclosure practices increases with their ownership interest in a firm. Kim et al. (2013) suggest that in an emerging market, controlling shareholders opportunistically use share repurchases to exploit their information advantage and secure control at the expense of firms' cash.

To sum up, these arguments suggest that investors, particularly in non-U.S. countries where repurchases are less popular, favor cash dividends over share repurchases. This effect is likely to be more pronounced during high-*EPU* periods, which are associated with increased agency problems, lending credence to our findings of stronger positive effect of *EPU* on cash dividend than on total payout.

Next, we examine whether dividend payers pay more dividends, and investors value dividend increases more highly, during times of policy uncertainty. In Model 8, we use a restricted sample that removes firm-year observations with zero dividends per share (*DPS*), and revisit the relation between dividend payout (*DPS*) and *EPU* for dividend payers.¹⁴ We find the coefficient of *EPU* is positive at the 5% significance level, similar to our findings in Model 6.

Moreover, we investigate the relation between market reactions to announcements of increases in dividends and economic policy uncertainty. In Model 9, we regress the cumulative abnormal return $[-1, +1]$ at the announcement of dividend increases (*CARDIVINCR*) on *EPU* and control variables. We find that *EPU* loads positively and significantly at the 5% level, suggesting that dividend increase decisions may serve as an effective means of mitigating investors' concerns about insiders' expropriation.

¹² We follow Faccio and Xu (2015), and re-estimate Model 1 after replacing the levels of the dependent and independent variables with their annual changes from $t-1$ to t . We report the results in Internet Appendix Table IA1. We find that changes in *EPU* are positively and significantly related to changes in *PAYOUT RATIO*.

¹³ We also use the ratio of cash dividends to net assets (total assets minus cash/short-term investment) as an alternative measure of payout policy. The results reported in Internet Appendix Table IA2 indicate that *EPU* remains significant at the 1% level.

¹⁴ In the same vein, we re-estimate Model 1 with the restricted sample, removing all zero-dividend firm-years, and find no material change in the coefficient on *EPU*. As shown in Internet Appendix Table IA3, it remains positive and significant at the 1% level.

Overall, during periods characterized by high levels of policy uncertainty, we find that firms tend to increase dividends.¹⁵ This evidence can be viewed through the lens of agency theory: When policy uncertainty increases, firms tend to reduce investment (Gulen and Ion 2016), and thus may distribute more cash to alleviate the agency costs of free cash flow. To support this channel, in unreported tests we find that EPU is associated with lower investments in our sample.¹⁶

As our results are qualitatively the same using each measure of corporate dividend payout, we next focus on *PAYOUT RATIO* and use Model 1 of Table 5 as our baseline model for further analysis.

3.2. Do the results reflect changes in the scaling variables?¹⁷

To further mitigate the concern that the main results are picking up the relation between the scaling variables and *EPU*, we examine the effects of *EPU* on the scaling variables. In Table 6, we rely on the main regression in Model 1 of Table 5 and replace *PAYOUT RATIO* with *\$IB* (net income before extraordinary items in \$US) in Model 1, *\$MV* (number of common shares outstanding multiplied by share price at fiscal-year end in \$US) in Model 2, *\$SALE* (total sales in \$US) in Model 3, and *\$CF* (net income before extraordinary items (IB) plus depreciation and amortization (DP) in \$US) in Model 4. The coefficients of *EPU* in these regressions are statistically insignificant at conventional levels.

We supplement this analysis by examining the effects of *EPU* on two variables constructed based on dividend per share (*DPS*). We construct the *DPS Initiation*, defined as a dummy variable that equals 1 if the firm has positive *DPS* in year *t* but 0 *DPS* in year *t*-1 and 0 otherwise, and *DPS Increase*, defined as a dummy variable that equals 1 if the firm has increased *DPS* in year *t* relative to year *t*-1, and 0 otherwise. We use the logistic regression of *DPS Initiation* and *DPS Increase* on the annual changes from *t*-1 to *t* of independent variables used in Model 1 of Table 5. Table 7 reports the results. We find that the estimated odds ratio of the changes in *EPU* is greater than 1 and significant at the 1% level in both regressions. This suggests that firms are more likely to initiate or increase dividend per share when economic policy uncertainty increases.

3.3. Do the results reflect potentially omitted variables?

To ease concerns that the omission of variables related to policy uncertainty and dividend payout may bias the estimates of our proxies for policy uncertainty, we include several additional control variables in Table 8.

In Model 1 of Table 8, we control for a more comprehensive set of firm-level characteristics. We introduce cash flows (*FCF*), defined as the sum of net income (IB) plus depreciation and amortization (DP) scaled by total assets (AT); *LEVERAGE*, computed as total long-term liabilities to total assets; and *MTB*, defined as the market value of assets (total book assets minus stockholder's equity plus market capitalization) divided by total book assets, to control for the degree of firms' financial constraints and growth prospects. In addition, Gulen and Ion (2016) find that, under policy uncertainty, firms may be concerned about investment irreversibility and restrict investments for precautionary reasons. We capture these concerns by including firm-level measures of capital expenditures (*CAPX*) and R&D expenditures (*XRD* and *XRDDUM*). *CAPX* and *XRD* are scaled by total assets (AT). *XRDDUM* equals 1 if R&D expenditures (*XRD*) are non-zero or non-missing, and 0 otherwise. Moreover, institutional shareholders with enhanced monitoring ability, unique information advantages, and possible tax preferences could affect firms' choice of dividend policy (Grinstein and Michaeli 2005). We capture the influence of institutional shareholders on dividend payout by percentage of shareholding by institutional investors (*IO*) from FactSet LionShares. Finally, following Kim et al. (2015), we add earnings uncertainty (*EARNVOL*) to capture the standard deviation of annual earnings over five years.

In Model 2, we consider additional industry- and country-level features that might correlate with corporate payout decisions. Prior studies suggest that industry-level and macroeconomic uncertainty could influence dividend payout policy (Floyd et al. 2015). To reduce concerns that the BBD index may capture the influence of other types of uncertainty on payout policy, we add controls for these variables. To measure industry-level uncertainty, we define *INDUSTRY_SHOCK* as the first principal component of the industry-year median of the absolute change in each of the seven economic shock variables (profitability, asset turnover, R&D, capital expenditures, employee growth, ROA, and sales growth) (Harford 2005). As a proxy for macroeconomic uncertainty, we follow Bonaime et al. (2018), and use the within-year standard deviation of firm sales growth in each country (*SD_SALES_GR*).

¹⁵ In addition to the news-based EPU index for 19 countries, BBD provide three other components of EPU for the U.S. to capture specific sources of policy uncertainty related to tax code changes and dispersion in the forecast of future monetary and fiscal policies. They compute the overall economic policy uncertainty index, *EPU (OVERALL)*, for the U.S. by averaging the four components with weights of 1/2 on the broad news-based policy uncertainty index, and 1/6 on the other three components. We use the overall BBD policy uncertainty index and its four individual components for a sample of U.S. firms from the quarterly Compustat files over the period from the first fiscal quarter of 1991 through the fourth fiscal quarter of 2015. Our U.S. quarterly data sample has 10,111 unique firms with 292,657 firm-quarter observations from 1991 to 2015. Applying the research design in our main analysis, we find that *EPU (OVERALL)* carries a positive sign at the 1% significance level, suggesting that firms pay more dividends during high-EPU periods. The analysis of the four subcomponents of EPU indicate that news-based and tax expiration components of policy uncertainty enter the regression positively and significantly at the 1% level, but the coefficients of the inflation disagreement and government purchases disagreement components are not statistically significant. These findings are similar to those of Gulen and Ion (2016) for an investment policy context. These results are reported in Internet Appendix Table IA4.

¹⁶ The coefficient estimate on EPU is negative and is statistically significant at the 1% level. These results are reported in Internet Appendix Table IA5.

¹⁷ We thank an anonymous reviewer for pointing out this concern.

Table 6
Relation between scaling variables and *EPU*.

Variables	(1)	(2)	(3)	(4)
	\$IB	\$MV	\$SALE	\$CF
<i>EPU</i>	−2.459 (−0.737)	10.579 (0.216)	16.184 (0.382)	2.536 (0.463)
Other controls	YES	YES	YES	YES
Observations	190,798	189,062	190,799	184,430
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Adj R-sq	0.716	0.835	0.896	0.833

This table presents results of regressing different scaling variables on *EPU* as well as firm- and country-level controls in Model 1 of Table 5. The dependent variables are *\$IB* (net income before extraordinary items in \$US) in Model 1, *\$MV* (number of common shares outstanding multiplied by share price at fiscal-year end in \$US) in Model 2, *\$SALE* (total sales in \$US) in Model 3, and *\$CF* (net income before extraordinary items (IB) plus depreciation and amortization (DP) in \$US) in Model 4, respectively. *EPU* is the natural logarithm of the average BBD (2016) policy uncertainty index over the 12-month period ending in the month of the fiscal year-end. All variables are defined in Table 1. Year dummies and firm fixed effects are included. All parameters are estimated using pooled OLS, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Table 7
Regressions with *DPS initiation* and *DPS increase* as dependent variables.

Variables	(1)	(2)
	<i>DPS Initiation</i>	<i>DPS Increase</i>
	odds ratio	odds ratio
ΔEPU	4.368*** (10.776)	1.311*** (8.324)
$\Delta SRVOL$	0.401*** (−2.946)	0.980 (−0.198)
ΔRE	1.020 (0.638)	1.049*** (2.645)
ΔTE	1.362 (0.983)	3.082*** (8.952)
ΔROA	3.945*** (4.873)	4.918*** (11.658)
ΔSGR	0.959 (−0.679)	0.975 (−0.994)
$\Delta SIZE$	1.189 (1.499)	2.946*** (21.712)
$\Delta CASH$	1.442 (1.317)	1.221* (1.848)
ΔGDP_PC	0.000*** (−16.393)	366.794*** (10.827)
Observations	37,585	117,391
Firm FE	YES	YES
Year FE	YES	YES
Pseudo R2	0.281	0.052

This table reports logistic regressions of *DPS Initiation* (Model 1) and *DPS Increase* (Model 2) on the annual changes of independent variables from *t*-1 to *t* used in Model 1 of Table 5 in the manuscript. *DPS Initiation* is defined as a dummy variable that equals 1 if the firm has positive *DPS* in year *t* but 0 *DPS* in year *t*-1, and 0 otherwise. *DPS Increase* is defined as a dummy variable that equals 1 if the firm has increased *DPS* in year *t* relative to year *t*-1, and 0 otherwise. Odds ratios are reported in the table. All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Gulen and Ion (2016) document a negative relation between the BBD index and GDP growth. We additionally introduce GDP growth (*GDPGROWTH*), defined as the annual percentage change in GDP, from WDI. Julio and Yook (2012) provide cross-country evidence that firms tend to reduce investment expenditures during election years. To ease concerns that our findings may be driven by the negative influence of elections on corporate investment or payout policy, we add an election year dummy variable (*ELECTION*) from the Database of Political Institutions. This variable equals 1 if there is an executive or legislative election in a given year, and 0 otherwise.

Table 8

EPU and dividend payout: alternative explanations.

Dependent variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Firm-level controls	Industry- and country-level controls	All	First Stage	Second Stage	First Stage	Second Stage
	PAYOUT RATIO	PAYOUT RATIO	PAYOUT RATIO	EPU	PAYOUT RATIO	EPU	PAYOUT RATIO
<i>EPU</i>	0.082*** (7.506)	0.080*** (8.211)	0.094*** (7.357)				
<i>EPU residual from first stage</i>					0.095*** (10.604)		0.094*** (7.296)
<i>CAPX</i>	−0.025 (−0.486)		0.014 (0.264)			−0.028 (−1.291)	
<i>XRD</i>	0.120** (2.060)		0.127** (2.086)			0.100*** (3.568)	
<i>XRDDUM</i>	0.047*** (4.868)		0.030*** (2.684)			0.026*** (4.769)	
<i>FCF</i>	0.236** (2.378)		0.308*** (3.035)			0.008 (0.193)	
<i>MTB</i>	−0.017*** (−9.643)		−0.014*** (−7.574)			−0.003*** (−2.985)	
<i>LEVERAGE</i>	−0.025 (−0.868)		0.002 (0.077)			−0.050*** (−3.985)	
<i>IO</i>	−0.062*** (−3.505)		−0.068*** (−3.823)			−0.008 (−1.064)	
<i>EARNVOL</i>	0.000 (0.790)		0.000 (0.669)	0.000* (1.912)		0.000 (0.238)	
<i>INDUSTRY_SHOCK</i>		−0.017*** (−6.637)	−0.021*** (−5.964)	−0.005*** (−4.203)		0.002 (1.382)	
<i>SD_SALES_GR</i>		−0.001*** (−8.668)	−0.001*** (−8.324)	−0.000*** (−14.555)		−0.000*** (−12.447)	
<i>GDPGROWTH</i>		−0.005*** (−3.599)	−0.006*** (−3.028)	−0.018*** (−37.631)		−0.023*** (−22.989)	
<i>ELECTION</i>		0.018*** (6.180)	0.026*** (6.538)	−0.030*** (−40.256)		−0.009*** (−7.133)	
<i>R_GDP_F</i>		−0.012*** (−8.109)	−0.008*** (−3.836)			−0.011*** (−9.501)	
<i>CLI</i>		−0.005** (−2.158)	−0.003 (−0.964)			−0.070*** (−39.751)	
<i>CCI</i>		−0.012*** (−5.088)	−0.008** (−2.333)			0.020*** (13.610)	
<i>TRUST</i>		1.016*** (12.393)	0.779*** (7.395)			−1.748*** (−26.241)	
<i>DIVTAX</i>		−0.001*** (−4.345)	−0.004*** (−7.961)			0.005*** (21.074)	
<i>CGTAX</i>		0.001 (1.293)	0.004*** (3.039)			−0.007*** (−17.456)	
<i>SHAREHOLDER PROTECT</i>		0.024*** (2.840)	0.048*** (3.622)			−0.230*** (−31.215)	
<i>CREDITOR PROTECT</i>		0.059*** (2.993)					
<i>SRVOL</i>	−0.299*** (−10.618)	−0.304*** (−15.132)	−0.275*** (−9.413)	0.105*** (14.356)	−0.335*** (−18.663)	0.158*** (15.136)	−0.297*** (−10.319)
<i>RE</i>	−0.005*** (−3.460)	−0.004*** (−4.448)	−0.005*** (−3.141)	−0.002*** (−3.451)	−0.007*** (−8.105)	0.001* (1.737)	−0.006*** (−3.896)
<i>TE</i>	0.200*** (7.474)	0.236*** (14.591)	0.243*** (8.849)	0.007 (1.216)	0.207*** (14.038)	−0.002 (−0.201)	0.224*** (9.216)
<i>ROA</i>	−0.262*** (−2.638)	−0.095*** (−8.590)	−0.341*** (−3.357)	−0.020*** (−4.544)	−0.086*** (−8.618)	0.001 (0.029)	−0.067*** (−4.395)
<i>SGR</i>	−0.032*** (−7.285)	−0.020*** (−7.429)	−0.027*** (−6.223)	−0.006*** (−5.678)	−0.029*** (−11.128)	−0.004*** (−2.536)	−0.035*** (−7.796)
<i>SIZE</i>	0.052*** (7.822)	0.046*** (11.964)	0.049*** (7.082)	0.018*** (12.303)	0.062*** (17.611)	0.012*** (3.636)	0.049*** (7.616)
<i>CASH</i>	0.003 (0.146)	0.006 (0.401)	0.020 (0.844)	0.001 (0.253)	0.037** (2.549)	−0.052*** (−6.074)	−0.039* (−1.651)
<i>GDPPC</i>	−0.038 (−1.371)	−0.355*** (−20.329)	−0.201*** (−5.496)	0.311*** (59.027)	−0.219*** (−15.737)	0.137*** (8.955)	0.008 (0.262)
Observations	85,755	144,343	77,545	169,973	169,973	77,545	77,545
Firm FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES

(continued on next page)

Table 8 (continued)

Dependent variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Firm-level controls	Industry- and country-level controls	All	First Stage	Second Stage	First Stage	Second Stage
	PAYOUT RATIO	PAYOUT RATIO	PAYOUT RATIO	EPU	PAYOUT RATIO	EPU	PAYOUT RATIO
Cluster by	firm	firm	firm	Firm	Firm	Firm	Firm
Adj R-sq	0.321	0.313	0.332	0.746	0.306	0.814	0.327

This table re-estimates Model 1 of Table 5 after adding potential omitted variables. The dependent variable is *PAYOUT RATIO* in Models 1–3, 5, and 7, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB), and is *EPU* in Models 4 and 6, defined as natural logarithm of the moving average of the monthly policy uncertainty index over the 12 months ending in the month of the fiscal year-end. *EPU_residual from first stage* in Model 5 (7) is the orthogonalized *EPU* against an expanded set of control variables in Model 4(6). The sample size varies across models due to the availability of the control variables added. All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Bloom et al. (2018) suggest that uncertainty is strongly countercyclical. As we stated earlier, an important predictor of firms' dividend policy is the perceived stability of future cash flows. These can be affected by the business cycles driven by economic uncertainty. Therefore, to address this concern, we follow Gulen and Ion (2016), and control for forecasted real GDP growth (*R_GDP_F*), the consumer confidence index (*CCI*), and composite leading indicators (*CLI*) from the WDI and OECD databases. Higher values for these variables indicate better market expectations of future economic conditions.

Guiso et al. (2008) argue that trust could influence investors' perceptions of the transaction and agency costs associated with stock market participation. To alleviate any concerns that our results capture a relation between corporate payout policy and investors' trust in companies and management, we control for trust (*TRUST*). We define *TRUST* as the percentage of survey responses from each country in response to the World Values Survey question, "Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?"

Prior literature further documents that, to maximize shareholder value, managers consider individual-level taxes when deciding corporate payout policies (Chetty and Saez 2005; Hanlon and Hoopes 2014). To ease concerns that changes in payout policy may capture shifts in tax rates of different countries during the same period, we follow Hail et al. (2017), and add the maximum statutory dividend tax rate (*DIVTAX*) and capital gains tax rate (*CGTAX*) for individuals (in percent). *DIVTAX* and *CGTAX* are time-varying until 2004, after which we use the 2004 tax rates.

Finally, La Porta et al. (2000) suggest that firms pay dividends as a result of strong shareholder protection. They find that minority shareholders have an incentive to pressure managers to distribute cash. In contrast, Brockman et al. (2014) find evidence that firms pay out higher dividends to overcome agency costs caused by weak legal institutional environments. They conclude that dividend payouts "dividend payouts serve as a substitute bonding mechanism when country-level legal protections fail" (p. 263). Faccio et al. (2001) find that poorly governed firms pay higher dividends. Brockman and Unlu (2009) similarly suggest that low dividend payouts can serve as a substitute for weak creditor rights. They find that, when creditor confidence in recovering claims during bankruptcy is low, creditors tend to use contractual covenants to restrict dividend payouts. These findings suggest that an increase in dividend payouts when facing policy uncertainty may reflect a country's degree of investor protection.

To mitigate this concern, we control for anti-director rights and its enforcement (*SHAREHOLDER PROTECT*), as well as for creditor rights and its enforcement (*CREDITOR PROTECT*). We construct *SHAREHOLDER PROTECT* as the first principal component of the revised anti-director rights index of Djankov et al. (2008b) and the law and order index of the International Country Risk Guide (ICRG). The time-invariant revised anti-director rights index measures the extent to which minority shareholders are protected in the corporate decision process in terms of the right to vote. The time-varying law and order index captures the strength and impartiality of the legal system and assesses popular observance of the law.

CREDITOR PROTECT is computed as the first principal component of the creditor rights index of Djankov et al. (2007), and the recovery rate for secured creditors. According to Djankov et al. (2007), the creditor rights index measures the four powers of secured lenders in bankruptcy granted by a country's laws and regulations. It thus ranges from 0 (weak creditor rights) to 4 (strong creditor rights). It is time-varying for 1990–2003 and is based on 2003 values thereafter. The recovery rate for secured creditors is the present value of the terminal value of the firm after bankruptcy costs, considering any violations of the absolute priority rule (see Djankov et al. 2008a). It ranges from 0 to 1.

In Model 3, we run a horse-race regression with a full set of firm-, industry-, and country-level controls.¹⁸ Each model in Table 8 shows that *EPU* remains a significant predictor of corporate dividend payouts, with significance at the 1% level. This suggests our main finding of a positive relation between economic policy uncertainty and dividend payouts is not driven by these omitted variables.

¹⁸ Including a full set of additional control variables in Model 3 reduces our sample size by 57% to 77,545 firm-year observations. Since we control for firm-fixed effects, *CREDITOR PROTECT* automatically drops in the regressions due to limited variation.

3.4. Do the results reflect other sources of uncertainty?

To rule out the concern that our measure of policy uncertainty may simply reflect general economic conditions or firm level uncertainty, we orthogonalize *EPU* against an expanded set of control variables, including firm-level volatility of sales and the macroeconomic variables used in Models 3 of Table 8. We report first- and second-stage results in Model 4 through 7. Importantly, we show that the orthogonalized *EPU* loads positively and significantly on our measure of dividend payout, lending further credence to our novel evidence that *EPU* is positively associated with dividend payout.¹⁹

The evidence above stresses that the documented effect of *EPU* on dividends is distinct from that of other sources of uncertainty, in particular, that of firm-level uncertainty. An important clarification, however, is in order. Taken on its face, the finding that firm-level uncertainty has an opposite effect on dividends compared to *EPU* may be surprising. But this result concurs with our theoretical discussion and predictions because *EPU* is inherently different from firm-level uncertainty. As discussed at the outset, while firm-specific uncertainty arises from factors unique to a firm (e.g., development of new products or management turnovers), *EPU* results from government policy and regulatory shocks and from other shocks that are largely beyond a manager's control. This makes *EPU* difficult to hedge for firms and investors. In contrast to firm-specific uncertainty, *EPU* shocks are largely exogenous and non-diversifiable (Duong et al. 2020), which may decrease the availability and attractiveness of investment opportunities. In turn, investors' preferences for cash dividends may increase in order to alleviate agency problems associated with *EPU*.

Compatible with this logic and building on insights from Izhakian and Yermack (2017), one can argue that policy-driven uncertainty is based on the concept of "Knightian uncertainty," as introduced by Knight (1921). This refers to "conditions under which the set of events that may occur are a priori unknown, and the odds of these possible events are also either not unique or are unknown" (Izhakian and Yermack 2017, p. 65).²⁰ Herron and Izhakian (2020) state that while with firm-level risk outcomes are a priori unknown but the odds of all possible outcomes are perfectly known, under Knightian uncertainty both outcomes and their odds are a priori unknown. They further argue that Knightian uncertainty leads ambiguity-averse investors to overweight the likelihoods of bad outcomes and underweight the likelihoods of good outcomes, making investments less attractive. This may increase investors' preference for cash payouts. Of particular relevance to the focus of our study, Herron and Izhakian (2020) find that dividend payout and the propensity to pay or to initiate a dividend decrease with firm-level risk and, conversely, increase with Knightian uncertainty.

3.5. Sample composition and alternative clustering

When examining summary statistics by country in Table 3, we observe that firms located in the U.S., Japan, and China account for 45%, 22%, and 11% of our main sample, respectively. To reduce concerns that observations from a single country are driving our results, in Models 1–3 of Table 9, we exclude firms from each of these countries in turn, and then re-estimate our main model (Model 1 of Table 5). To check the stability of results over time, we split the sample into two equal subperiods: 1991–2003 and 2004–2015. We report results in Models 4 and 5. In Model 6, we use a weighted regression with weights equal to the inverse of the number of firm-year observations in each country to account for the variation in observations across countries. We also consider alternative ways to cluster the standard errors. We adjust standard errors for clustering at the country level in Model 7 and use two-way clustering at both the country-year and firm level in Model 8, following Faccio and Xu (2015). We find that *EPU* loads significantly positively at the 1% level in Models 1–8, suggesting that our previous findings are not affected by overrepresented countries, different sample periods, or by alternative more complex underlying error structures.²¹

3.6. Endogeneity

A natural concern with our empirical analysis is potential endogeneity stemming from omitted variables, measurement errors, and simultaneity, which may bias the estimated coefficient on *EPU* (Roberts and Whited 2013). To address these concerns, we control for various firm- and country-level characteristics, in addition to firm and year fixed effects in the analyses above. Moreover, to address concerns that, in determining payout ratios, the BBD index may capture other factors that are potentially correlated with *EPU*, in Table 8, we include proxies for corporate financial soundness, growth prospects, corporate investment, institutional shareholding, uncertainty type, macroeconomic conditions, trust, investor-level taxes, and legal institutions.

While it is unlikely that firm-level payout decisions contribute to policy-related uncertainty at a macro level, it is possible that some unobserved factors may affect policy uncertainty and corporate policy payout decisions simultaneously (i.e., $\text{cov}(EPU, \varepsilon_{i,t}) \neq 0$). The OLS estimate in our main regression could understate the true effect of policy uncertainty on dividend policies if the unobserved factors affect *EPU* and dividend payouts in the opposite direction or overstate the true effect if they move both *EPU* and dividends in the same direction (Jiang 2017).

¹⁹ In a related study, Bonaime et al. (2018) show that the effect of policy-specific uncertainty on M&A activity is not absorbed by firm-level volatility.

²⁰ Stated differently, it builds on the lack of any quantifiable knowledge about some possible occurrence, as opposed to the presence of quantifiable risk. Interestingly, this uncertainty is distinctly different from firm-specific volatility that typically treats the set of future outcomes (such as the range of future earnings or stock prices) as known, with certain probabilities assigned to them and adding to one.

²¹ Although the problem of too few clusters (Cameron and Miller 2015) makes exact inferences difficult, these robustness checks provide a sense of the potential relative impact of a more complex cross-firm, within-country correlated error structure.

Table 9

EPU and dividend payout: sample composition and alternative clustering.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Exclude U. S.	Exclude Japan	Exclude China	Period 1991–2003	Period 2004–2015	Weighted Least Squares	Cluster by Country	Two-way Clustering
<i>EPU</i>	0.106*** (10.995)	0.034*** (3.766)	0.086*** (8.790)	0.097*** (6.321)	0.105*** (10.757)	0.136*** (12.672)	0.103*** (3.163)	0.103*** (3.439)
<i>SRVOL</i>	−0.513*** (−15.230)	−0.218*** (−13.478)	−0.274*** (−14.731)	−0.145*** (−6.903)	−0.362*** (−13.640)	−0.266*** (−15.826)	−0.331** (−2.714)	−0.331*** (−7.229)
<i>RE</i>	−0.000 (−0.063)	−0.003*** (−3.776)	−0.005*** (−6.064)	−0.005*** (−4.217)	−0.006*** (−4.065)	−0.005*** (−5.636)	−0.007* (−2.078)	−0.007*** (−5.074)
<i>TE</i>	0.426*** (15.628)	0.163*** (11.683)	0.180*** (11.750)	0.182*** (9.904)	0.279*** (13.470)	0.139*** (9.340)	0.201** (2.244)	0.201*** (7.787)
<i>ROA</i>	−0.360*** (−9.117)	−0.033*** (−3.622)	−0.088*** (−8.866)	−0.058*** (−5.837)	−0.107*** (−5.779)	−0.059*** (−6.691)	−0.090 (−1.153)	−0.090*** (−4.455)
<i>SGR</i>	−0.049*** (−7.884)	−0.017*** (−7.582)	−0.031*** (−11.950)	−0.016*** (−6.123)	−0.030*** (−7.265)	−0.020*** (−10.132)	−0.029** (−2.153)	−0.029*** (−6.637)
<i>SIZE</i>	0.114*** (17.667)	0.032*** (9.831)	0.058*** (16.289)	0.036*** (7.991)	0.067*** (11.733)	0.046*** (13.671)	0.063* (2.052)	0.063*** (7.515)
<i>CASH</i>	−0.067** (−2.429)	0.039*** (3.032)	0.037** (2.488)	−0.028* (−1.674)	−0.016 (−0.814)	0.050*** (3.686)	0.028 (0.688)	0.028 (1.239)
<i>GDP_PC</i>	−0.251*** (−15.469)	−0.312*** (−21.580)	0.096*** (2.617)	0.174* (1.738)	−0.261*** (−15.959)	−0.199*** (−12.470)	−0.236*** (−4.164)	−0.236*** (−3.320)
Observations	98,996	139,084	160,262	77,299	101,887	179,186	179,186	179,186
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Cluster by	Firm	Firm	Firm	firm	firm	Firm	Country	Firm & Country-year
Adj R-sq	0.239	0.362	0.310	0.386	0.324	0.287	0.304	0.304

This table re-estimates Model 1 in Table 5 using different subsamples in Models 1 to 5, weighted least squares (WLS) in Model 6, and alternative clustering of the standard errors in Models 7 and 8. The dependent variable is *PAYOUT_RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm in Models 1–6. Standard errors are clustered by country in Model 7 and by two dimensions (country-year and firm) in Model 8. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

It is unclear how the unobserved factors influence policy uncertainty and dividend policies. As in other empirical analyses, we are unable to randomly assign the level of policy uncertainty to firms or to observe their corresponding changes in dividend payouts. In this section, therefore, we use the instrumental variables approach and a placebo test to further ease simultaneity concerns. Our approaches to addressing endogeneity, including the choice of instruments, follow recent EPU studies. As summarized in Table IA6, among the ten studies that address endogeneity using the IV approach, six use political polarization as an instrument.

A good instrumental variable in our setting should affect economic policy uncertainty in a significant way (relevance condition) but should not affect the dividend payout except indirectly through its impact on policy uncertainty (exclusion condition). Gulen and Ion (2016) quantify the degree of political polarization in the United States Senate as an instrument for the level of *EPU*. Relying on political science literature, they focus on the first dimension of the DW-NOMINATE scores of McCarty et al. (1997) that are designed to capture legislators' positions on government intervention in the economy over time. Specifically, they measure the level of political polarization with the difference between the average DW-NOMINATE scores for the Republican and Democratic party members in the Senate.

In the same vein, we draw on the Database on Political Institutions for our international sample. We propose a plausibly exogenous polarization measure (*POLAR*) as the product of political fractionalization and government opposition. We define political fractionalization as the probability that two deputies picked at random from the legislature will be of different parties, and government opposition as 1 minus the government share of seats in parliament (government support). A highly divided legislature would likely increase the difficulty of reaching a consensus on any given policy, which could lead to greater EPU.

Grier et al. (2015) find strong evidence that political fractionalization is related to longer delays in fiscal policy reforms and stabilization of high deficits. McCarty (2004) argues that partisan polarization makes it more difficult to build legislative coalitions, which can lead to policy gridlock, hinder the efficient functioning of the administrative state, and cause greater variations in policy outcomes. Taken together, our interaction instrument uses a double-whammy scenario against smooth policymaking. It is expected to have a significantly positive relation with *EPU* levels, meeting the relevance condition. In addition, it is less likely that the level of political polarization (*POLAR*) will affect corporate dividend policies through channels other than its effect on *EPU*. This satisfies the exclusion restriction.

POLAR varies from 0 to 1, with larger values indicating a more divided legislature. We report first- and second-stage results of this analysis in Models 1 and 2 of Table 10. In Model 1, we find that *POLAR* loads positively at the 1% level. This suggests that EPU is higher in the presence of a more divided legislature, consistent with the literature. Model 2 reports the second-stage results, using the fitted values of *EPU* computed from the first stage. We find that the coefficient estimate of *EPU* is positive and significant at the 1% level,

alleviating the concern that our previous findings are driven by endogeneity.

The magnitude of the *EPU* estimate in the IV regression is six times that of the uninstrumented estimate. To alleviate weak instrument concerns, we conduct an *F*-test of the excluded exogenous variable in the first-stage regression, in which the null hypothesis is that the instrument is weak. The Kleibergen–Paap Wald *rk F* statistic²² passes the Stock and Yogo (2005) test with a maximum size of 10%, indicating that *POLAR* captures the variation in *EPU*.

Next, we check rank by using the underidentification test of the Kleibergen–Paap *rk LM* statistic, in which the null hypothesis is that the matrix of reduced-form coefficients on the excluded instruments is not full rank. The chi-square value rejecting the null at the 1% is 790.57, which suggests the model is well identified at the 1% level.

To complement the instrumental variables approach, we run placebo tests to reduce endogeneity concerns that the time-varying omitted variables may simultaneously drive both policy uncertainty and dividend payouts. We randomly draw from the sample distribution of *EPU* within each country to construct the 1000 fake economic policy uncertainty indices, $\widetilde{EPU}$, over our sample period. We replace *EPU* with $\widetilde{EPU}$, and re-estimate the baseline regression model. If the effect of *EPU* on dividends merely reflect time trends in dividend payouts driven by the omitted variables, we expect that $\widetilde{EPU}$ will continue to load positively and significantly.

Model 3 of Table 10 reports the average of the coefficient estimates and *t*-statistics for each independent variable over 1000 regressions. We find the average coefficient on $\widetilde{EPU}$ is -0.0005 and statistically insignificant. More importantly, none of the 1000 coefficients on $\widetilde{EPU}$ is greater than the true coefficient on *EPU* (0.103), indicating that our main results survive the placebo test.

4. Additional analyses

The empirical analyses in previous sections suggest a robust and positive relation between *EPU* and dividend payouts. We interpret our results as consistent with the free cash flow hypothesis rather than with the precautionary motives hypothesis. Managers have an incentive to retain firm earnings in order to overinvest and expropriate rents (Jensen 1986). This problem may be aggravated by declining corporate investment (Gulen and Ion 2016). We thus expect policy uncertainty to have a stronger effect on firms that are prone to severe agency problems.

In this section, we explore these explanations further. We perform split-sample tests to examine how the effect of *EPU* on dividend payouts is affected by growth opportunities, positive free cash flows, strength of the legal protection of shareholders and creditors, and governance quality. We split the sample according to the mean values of the continuous conditioning variables for subsample tests below (0 and 1 for binary conditioning variables).

4.1. The role of growth opportunities and free cash flow

Firms with fewer growth opportunities and higher free cash flows tend to have marked agency problems (Chen et al. 2011; Lang et al. 1991). We capture firms' growth opportunities by using sales growth (*SGR*), as defined above.²³ We split the sample by the mean level of *SGR*. The results are in Models 1 and 2 of Table 11. We find that the coefficient on *EPU* loads positively at the 5% level or better in both sets of subsamples but it has greater influence on firms with fewer growth opportunities.

Next, we follow Brockman and Unlu (2009), and measure free cash flows using a binary variable (*POSFCF*) that equals 1 if the firm has positive free cash flows and 0 otherwise. Free cash flow is defined as net income (*IB*) plus depreciation and amortization (*DP*) scaled by total assets (*AT*). In Models 3 and 4, we divide the sample by whether the firm has positive free cash flows (*POSFCF* = 1 or 0). The results show that the positive relation between *EPU* and *PAYOUT RATIO* is statistically and economically more significant for firms with positive free cash flows. This suggests that firms that are more vulnerable to agency problems (few growth opportunities and positive free cash flows) pay out more cash when policy uncertainty is higher.

4.2. The role of legal institutions

The level of investor protection determines the extent of agency problems and affects dividend payouts (e.g., La Porta et al. 2000; Faccio et al. 2001). In Table 12, we test whether the strength of minority shareholder protection influences the positive effect of *EPU* on payout ratios.

Our first measure of the strength of shareholder protection is the composite *SHAREHOLDER PROTECT* index, as defined in Section 3.3. We split the sample along the mean of *SHAREHOLDER PROTECT*. Results for the above- and below-mean subsamples are in Models 1 and 2, respectively. We find that the coefficients on *EPU* are positive and significant at the 1% level in both models, but economically more important in the below-mean sample. These findings suggest that the positive relation between *EPU* and *PAYOUT RATIO* is more pronounced in countries with weaker shareholder protection.

Following La Porta et al. (2006) and Djankov et al. (2008b), we next measure the strength of investor protection using 1) *DISCLOSE*,

²² Andrews et al. (2019) suggest that in the setting with a single endogenous variable, the Kleibergen–Paap Wald *rk F* statistic is equivalent to a non-homoskedasticity-robust *F*-statistic proposed by Olea and Pflueger (2013) for testing weak identification.

²³ We also measure growth opportunities using *TobinQ*. We compute this variable as the ratio of the market value of assets to total book assets, where the market value of assets equals total book assets minus stockholders' equity plus market capitalization. The results are similar to the findings for *SGR*. We report the findings in Internet Appendix Table IA7.

Table 10
Endogeneity.

Variables	(1)	(2)	(3)
	IV regression		Placebo Test
	First (DV: EPU)	Second (DV: Payout Ratio)	
<i>POLAR</i>	0.679*** (30.434)		
Fitted <i>EPU</i>		0.630*** (9.386)	
$\widehat{EPU}$			−0.000 (−0.210)
<i>SRVOL</i>	0.077*** (11.299)	−0.369*** (−19.970)	−0.324*** (−18.710)
<i>RE</i>	−0.001** (−2.115)	−0.006*** (−6.600)	−0.007*** (−8.506)
<i>TE</i>	0.027*** (4.672)	0.197*** (13.338)	0.201*** (14.009)
<i>ROA</i>	−0.028*** (−6.369)	−0.075*** (−7.458)	−0.092*** (−9.381)
<i>SGR</i>	−0.005*** (−5.407)	−0.023*** (−9.425)	−0.029*** (−12.208)
<i>SIZE</i>	0.013*** (9.747)	0.049*** (14.069)	0.065*** (19.321)
<i>CASH</i>	−0.014*** (−2.697)	0.031** (2.209)	0.027** (1.960)
<i>GDP_PC</i>	0.310*** (58.983)	−0.413*** (−17.401)	−0.206*** (−14.965)
Observations	173,201	173,201	179,186
Firm FE	YES	YES	YES
Year FE	YES	YES	YES
Kleibergen-Paap Wald rk F statistic	926.26		
Kleibergen-Paap rk LM test	790.57***		

This table reports results from an instrumental variables regression of *PAYOUT RATIO* on *EPU*, as well as firm- and country-level controls and a placebo test. The dependent variable is *PAYOUT RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). The instrumental variable is *POLAR*, the product of probability, that two deputies picked at random from the legislature will be of different parties, and government opposition constructed as 1 minus government support (or the government share of seats in parliament). All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. The coefficient and *t*-statistics reported in Model 3 are the average of coefficient and *t*-statistics from regressions using 1000 randomized $\widehat{EPU}$ s. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

the arithmetic mean of six indicators of prospectus disclosure requirements (i.e., prospects, compensation, shareholders, inside ownership, irregular contracts outside the ordinary course of business, and transactions with related parties); 2) *SECURITY REG*, the first principal component of the disclosure index, the liability standard index, and the public enforcement index, which measure disclosure requirements and liability standards that facilitate private enforcement as well as the strength of public enforcement by supervisory bodies; and 3) *PUBLIC ENFORCE*, the index of public enforcement, including fines and prison terms for related parties in self-dealing if all disclosure and approval requirements have been met.

We split the sample along the mean levels of *DISCLOSE*, *SECURITY_REG*, and *PUBLIC ENFORCE*, and the results are in Models 3–8. Note that *EPU* loads positively at the 5% level or better in all subsamples (except in Models 5 and 7), and exhibits economically larger coefficients in all three below-mean subsamples. These results suggest that, during times of greater policy uncertainty, shareholders demand (and managers consent to) the distribution of cash as a substitute for formal investor protection in countries where security laws provide insufficient protection.

Brockman and Unlu (2009) further show that managers in countries with weak creditor rights are under greater pressure to lower dividend payouts because of creditors' concerns about debt repayments. That is, in the presence of weak creditor rights, a more restrictive payout policy can minimize the firm's agency costs of debt (Brockman and Unlu 2009).

Drawing on these insights, we argue that managers need to employ relatively more restrictive dividend policies to reassure weakly protected creditors during times of high policy uncertainty. To capture the extent of creditor rights and debt enforcement, we use the variable *CREDITOR PROTECT* as defined in Section 3.3. We then divide the sample by the mean value of *CREDITOR PROTECT*, and the results are in Models 9 and 10. While *EPU* continues to load significantly positively in both subsamples, its effect appears to be more pronounced in countries with higher creditor rights. This may be because the substitution effect between a restrictive payout policy and poor creditor rights (Brockman and Unlu 2009) is more pronounced in countries with weak creditor rights, possibly in an effort to minimize firms' agency costs of debt during high-*EPU* periods.

Table 11

EPU and dividend payout: the role of growth opportunities and free cash flow.

Variables	(1)	(2)	(3)	(4)
	<i>SGR</i>		<i>POSFCF</i>	
	High	Low	YES	NO
<i>EPU</i>	0.025** (2.109)	0.145*** (11.889)	0.107*** (11.794)	0.036* (1.809)
<i>SRVOL</i>	−0.208*** (−10.099)	−0.392*** (−14.290)	−0.386*** (−16.749)	−0.033*** (−2.756)
<i>RE</i>	−0.004*** (−4.863)	−0.009*** (−7.158)	−0.001 (−0.518)	−0.001 (−0.910)
<i>TE</i>	0.112*** (7.615)	0.275*** (11.987)	0.241*** (13.584)	0.032** (2.475)
<i>ROA</i>	−0.040*** (−3.470)	−0.107*** (−6.980)	−0.551*** (−13.940)	0.004 (0.762)
<i>SGR</i>	−0.012*** (−3.819)	−0.041*** (−5.624)	−0.051*** (−10.489)	−0.002 (−1.101)
<i>SIZE</i>	0.032*** (9.571)	0.088*** (15.088)	0.058*** (14.413)	0.003 (0.803)
<i>CASH</i>	0.015 (1.157)	0.039 (1.638)	0.062*** (3.430)	0.000 (0.024)
<i>GDP_PC</i>	−0.317*** (−17.934)	−0.159*** (−8.123)	−0.269*** (−18.300)	−0.030 (−1.185)
Observations	75,072	104,114	148,659	23,700
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Adj R-sq	0.405	0.262	0.300	0.677
Difference in the coefficient on EPU (High/YES – Low/NO)	−0.120*** (−7.071)		0.071*** (3.237)	

This table presents estimation results for various subsamples of regressions of *PAYOUT RATIO* on measures of policy uncertainty (*EPU*) and firm- and country-level controls. The dependent variable is *PAYOUT RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). *EPU* is the natural logarithm of the average BBD (2016) policy uncertainty index over the 12-month period ending in the month of the fiscal year-end. We divide the sample into above- and below-mean subsamples according to the mean values (or 1/0 values) of the partitioning variables, namely, *SGR* (Models 1 and 2) and *POSFCF* (Models 3 and 4). All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

4.3. The role of ownership structure

Farre-Mensa et al. (2014), in their review of the extant literature on dividend policy, argue that, among traditional motives for payouts, agency theory has the most robust empirical support. Indeed, dividends can play a role in mitigating agency costs and limiting expropriation by removing corporate wealth from the control of self-interested insiders (Rozeff 1982; Jensen 1986; Faccio et al. 2001). This is consistent with DeAngelo et al.'s (2006) evidence that firms pay dividends to mitigate agency problems. Faccio et al. (2001) also show that that significantly higher dividends are paid by corporations vulnerable to insider expropriation (i.e., “tightly affiliated” to a business group via a chain of control that comprises a lower ownership/control ratio). This suggests that higher dividends are paid to allay investors’ concerns of potential expropriation by controlling shareholders.

Similarly, Pindado et al. (2012) provide evidence from nine Eurozone countries that suggests that family firms distribute higher and more stable dividends to alleviate expropriation concerns of minority investors. Berzins et al. (2018) show that controlling shareholders may mitigate serious agency conflicts by choosing more minority-friendly dividend policies. This is consistent with their 2019 findings that potential conflicts between majority and minority shareholders strongly influence how dividends respond to taxes (Berzins et al. 2019).

Furthermore, Abreu and Gulamhussen (2013) find that even with stringent regulatory pressure, firms disgorge more dividends to accommodate the increased need for monitoring. Officer (2011) shows that firms with “good governance” tend to pay lower dividends. Grinstein and Michaely (2005) also document that firms with high institutional holdings tend to be dividend payers, but they pay lower dividends.

Drawing on these insights, we investigate the extent to which agency costs embedded in ownership structure condition our evidence of the positive link between EPU and dividends.

Ownership structure is at the core of agency problems (e.g., Shleifer and Vishny 1997). We focus on the role of large shareholders in the positive relation between EPU and dividend payout, using ownership information from Osiris (Bureau van Dijk) over the January 2004–December 2016 period. We extract two dummy variables: *BLOCKHOLDER*, which equals 1 if the firm has a blockholder who holds more than 25%, and 0 otherwise, and *FAMILY*, which equals 1 if the type of blockholder is “individual(s)/family(ies),” and 0 otherwise. Panel A of Table 13 presents the results of subsamples split by whether firms are owned by blockholders in Models 1 and 2, or families in Models 3 and 4, respectively. The coefficients of *EPU* are positive and significant across all four models but show no significant difference between closely or widely held firms, or between family and non-family firms.

Table 12

EPU and dividend payout: the role of legal institutions.

Partitioning variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	<i>SHAREHOLDER_PROTECT</i>		<i>DISCLOSE</i>		<i>SECURITY_REG</i>		<i>PUBLIC_ENFORCE</i>		<i>CREDITOR_PROTECT</i>	
	HIGH	LOW	HIGH	LOW	HIGH	LOW	HIGH	LOW	HIGH	LOW
<i>EPU</i>	0.090*** (7.724)	0.121*** (10.864)	0.027** (2.032)	0.097*** (7.109)	0.027 (1.639)	0.083*** (6.910)	−0.006 (−0.400)	0.128*** (12.381)	0.096*** (6.255)	0.050*** (4.707)
<i>SRVOL</i>	−0.270*** (−12.150)	−0.252*** (−9.407)	−0.159*** (−9.383)	−0.475*** (−9.440)	−0.173*** (−10.035)	−0.403*** (−9.021)	−0.265*** (−5.415)	−0.344*** (−18.593)	−0.429*** (−8.155)	−0.225*** (−13.916)
<i>RE</i>	−0.006*** (−5.563)	−0.004** (−2.508)	−0.001 (−1.288)	0.028*** (4.373)	−0.001 (−1.202)	0.029*** (4.561)	0.005 (1.318)	−0.008*** (−8.706)	0.030*** (3.990)	−0.002* (−2.482)
<i>TE</i>	0.256*** (13.328)	0.204*** (9.506)	0.110*** (7.299)	0.498*** (12.110)	0.108*** (7.036)	0.469*** (12.449)	0.170*** (4.163)	0.201*** (13.112)	0.570*** (12.808)	0.138*** (9.739)
<i>ROA</i>	−0.135*** (−10.697)	−0.033* (−1.923)	−0.030*** (−3.582)	−0.629*** (−8.891)	−0.025*** (−3.002)	−0.628*** (−9.417)	−0.029 (−0.798)	−0.095*** (−9.443)	−0.909*** (−10.118)	−0.018** (−2.144)
<i>SGR</i>	−0.032*** (−9.996)	−0.009*** (−2.732)	−0.015*** (−7.131)	−0.099*** (−7.912)	−0.014*** (−6.707)	−0.086*** (−7.813)	−0.042*** (−4.571)	−0.025*** (−10.748)	−0.106*** (−7.385)	−0.012*** (−5.957)
<i>SIZE</i>	0.071*** (15.478)	0.034*** (5.621)	0.024*** (6.743)	0.143*** (14.417)	0.022*** (6.125)	0.137*** (14.812)	0.034*** (3.260)	0.064*** (17.926)	0.157*** (14.091)	0.023*** (6.908)
<i>CASH</i>	−0.034* (−1.861)	0.046** (2.281)	0.066*** (4.818)	−0.095** (−2.140)	0.067*** (4.822)	−0.088** (−2.222)	0.142*** (3.408)	0.016 (1.082)	−0.060 (−1.228)	0.042*** (3.230)
<i>GDP_PC</i>	0.279*** (3.972)	−0.297*** (−16.986)	−0.194*** (−4.973)	−0.059 (−0.862)	0.291 (1.557)	−0.129*** (−3.308)	−0.425*** (−7.783)	−0.239*** (−16.652)	−0.152** (−2.003)	−0.313*** (−20.955)
Observations	115,411	63,775	93,586	66,574	85,372	74,788	30,432	148,754	59,243	111,729
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj R-sq	0.307	0.331	0.323	0.187	0.323	0.194	0.350	0.287	0.185	0.329
Difference in the coefficient on <i>EPU</i> (High – Low)	−0.031* (−1.904)		−0.070*** (−3.664)		−0.056*** (−2.691)		−0.134*** (−7.102)		0.046** (2.473)	

This table presents estimation results for various subsamples for regressions of *PAYOUT RATIO* on measures of policy uncertainty (*EPU*) and firm- and country-level controls. The dependent variable is *PAYOUT RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). *EPU* is the natural logarithm of the average BBD (2016) policy uncertainty index over the 12-month period ending in the month of the fiscal year-end. We divide the sample into above- and below-mean subsamples according to the mean values of partitioning variables, namely *SHAREHOLDER PROTECT* (Models 1 and 2), *DISCLOSE* (Models 3 and 4), *SECURITY REG* (Models 5 and 6), *PUBLIC ENFORCE* (Models 7 and 8), and *CREDITOR PROTECT* (Models 9 and 10). All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm-fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

To reduce the concerns that U.S. firms dominate the sample and may bias the results, we re-run Models 1–4 using a sample that excludes U.S. firms. The results are in Models 5–8, respectively. We report similar results. Firms held by blockholders and family-owned firms pay more dividends in response to higher uncertainty in economic policy, although the differences are statistically insignificant.

Note that we should be cautious in claiming that ownership structure is not relevant in moderating the effect of EPU on dividends. This is because we have an international sample, and different ownership structures may be associated with different agency roles under different institutional contexts.²⁴ Accordingly, to shed further light on the role of agency problems associated with ownership concentration, we sort our closely held/family-owned subsamples based on the level of shareholder protection. All else equal, ownership concentration may be associated with more agency problems (e.g., between large shareholders and minority interest) in economies with weak shareholder protection. The results are in Panel B of Table 13.

Interestingly, we find that, in economies with weaker shareholder protection, firms with large blockholding and family ownership distribute more dividends during high-EPU periods than widely held firms and non-family firms, respectively. This suggests that, during periods with increased uncertainty over government policies, dividend payout plays a relevant role in alleviating the agency problems associated with concentrated ownership.

4.4. The role of board characteristics

Our findings above have shown that the role of dividend payouts becomes more significant as agency problems intensify due to higher EPU. Next, we examine how the positive relation between policy uncertainty and dividend payouts is affected by the strength of corporate governance at the firm level, as captured by the attributes of the board of directors. Governance experts and shareholder activists generally recommend separating the CEO and chairman roles (Larcker and Tayan 2016). Prior literature has further suggested that board ownership (Morck et al. 1988), independent directors (Black and Kim 2012; Core et al. 1999; Nguyen and Nielsen 2010), and blockholders (Shleifer and Vishny 1997) are associated with improved monitoring.

Turning to the role of dividends, Easterbrook (1984) and Rozeff (1982) argue that dividends serve as a corporate governance device that reduces free cash flow problems and exposes firms to market monitoring when raising funds through the market.

We obtain information on corporate governance from NRG Metrics Data, a database used by listed firms and institutional investors. NRG Metrics employs expert analysts to manually review and collect detailed information on the characteristics of CEOs, boards, and ownership from corporate annual reports dating back to 2007.²⁵ We capture corporate governance quality using the following variables: 1) *INDEP_CHAIRMAN*, a binary variable that indicates whether the board chairman is independent; 2) *BOARD_OWNERSHIP*, the percentage of shareholding by all board members; 3) *BOARD_INDEP*, the percentage of independent members on the board; and 4) *EXTERNALBLOCK*, the percentage of shareholding by all external blockholders. The results are in Table 14.

In Models 1 and 2, we split the sample along independence of the firm's chairman. The coefficient on *EPU* is positive and highly significant at the 1% level only when the chairman is not independent. This suggests that the pressure to pay out cash is higher in firms with severe agency problems. In Models 3 and 4 (5 and 6, 7 and 8), we split the sample along the mean level of *BOARD_OWNERSHIP* (*BOARD_INDEP*, *EXTERNALBLOCK*). While the estimated coefficients on *EPU* are statistically insignificant in the above-mean subsamples, they are positive and significant at the 5% level or better in the below-mean subsamples. This suggests that *EPU* exerts stronger pressure on *PAYOUT_RATIO* when firms have insufficient interest alignment (a lower level of board ownership) and weaker monitoring (reflected by a smaller percentage of independent directors on the board, as well as by lower shareholdings by external blockholders).²⁶

However, the shorter sample period and limited firm coverage of NRG Metrics Data reduce our sample size dramatically (a 94% loss of observations). To alleviate this concern, we supplement NRG Metrics with ISS, BroadEx, and ASSET4 to collect the percentage of independent directors (*BOARD_INDEP_EXPAND*) and construct a sample of 49,918 observations across countries. In Models 9 and 10, we split the sample by the mean level of *BOARD_INDEP_EXPAND* and find similar results as in Models 5 and 6, reducing the concern that our previous findings on the role of board characteristics is driven by the small sample size of NRG Metrics.

Taken together, our results on the role of corporate governance variables (e.g., free cash flow, ownership structure, and board attributes) in altering the link between EPU and dividends lend further credence to the agency theory motive of corporate payout policy.

5. Conclusion

In this paper, we examine the effect of economic policy uncertainty on corporate dividend policy. If the precautionary motive dominates, managers should pay out less cash during times of greater policy-related uncertainty. Conversely, if EPU is associated with

²⁴ For example, concentrated ownership structure (i.e., large shareholders) may play an important monitoring role in some countries/institutional contexts or could be associated with more agency problems (see Attig et al. 2016, for a discussion).

²⁵ The NRG Metrics data provide good coverage for large-cap indices from developed and advanced emerging countries, middle- and small-cap indices for European countries, and countries that provide annual reports in English (India, Indonesia, Philippines, and Russia). In our sample, Chile is not captured by NRG Metrics Data.

²⁶ To some extent, our findings of a more pronounced impact of EPU on dividends in firms that are more prone to agency problems corroborate Faccio et al.'s (2001) evidence that higher dividends are paid to offset investors' concerns of greater expropriation risk.

Table 13

EPU and dividend payout: the role of ownership structure.

Panel A. The impact of EPU on dividend payout in closely held (family) firms vs. non-closely held (non-family) firms								
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Full Sample				Excluding U.S. Firms			
	Blockholder		Family		Blockholder		Family	
	YES	NO	YES	NO	YES	NO	YES	NO
<i>EPU</i>	0.122*** (6.387)	0.116*** (8.948)	0.151*** (4.583)	0.112*** (10.367)	0.126*** (6.271)	0.097*** (6.583)	0.157*** (4.051)	0.098*** (8.021)
<i>SRVOL</i>	−0.178*** (−2.966)	−0.263*** (−7.503)	−0.132 (−1.507)	−0.256*** (−8.025)	−0.179** (−2.566)	−0.319*** (−6.034)	−0.015 (−0.104)	−0.314*** (−7.145)
<i>RE</i>	−0.002 (−0.659)	−0.005*** (−2.654)	0.002 (0.278)	−0.006*** (−2.905)	0.007 (1.206)	0.018** (2.494)	0.081** (2.481)	0.013** (2.430)
<i>TE</i>	0.332*** (5.651)	0.240*** (8.713)	0.142* (1.686)	0.288*** (11.145)	0.289*** (4.562)	0.431*** (9.403)	0.149 (1.197)	0.420*** (10.961)
<i>ROA</i>	−0.161** (−2.381)	−0.092*** (−3.852)	−0.087 (−0.697)	−0.120*** (−5.587)	−0.315** (−2.557)	−0.489*** (−5.962)	−0.244 (−0.790)	−0.453*** (−6.740)
<i>SGR</i>	−0.050*** (−4.789)	−0.036*** (−6.177)	−0.011 (−0.549)	−0.040*** (−7.651)	−0.056*** (−4.386)	−0.058*** (−5.091)	−0.044 (−1.582)	−0.055*** (−6.400)
<i>SIZE</i>	0.088*** (5.533)	0.068*** (8.211)	0.037 (1.219)	0.078*** (10.494)	0.082*** (4.908)	0.119*** (9.112)	0.029 (0.615)	0.114*** (11.109)
<i>CASH</i>	−0.039 (−0.743)	−0.013 (−0.505)	0.149* (1.723)	−0.040* (−1.699)	−0.058 (−1.169)	−0.151*** (−3.579)	−0.033 (−0.336)	−0.129*** (−4.006)
<i>GDP_PC</i>	0.315*** (5.890)	0.195*** (4.932)	0.467*** (3.293)	0.263*** (9.352)	0.358*** (6.299)	0.252*** (5.279)	0.529*** (2.866)	0.333*** (9.610)
Observations	21,328	69,730	8693	82,365	18,607	44,537	5217	57,927
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Adj R-sq	0.396	0.317	0.360	0.325	0.394	0.261	0.388	0.286
Difference in the coefficient on Policy Uncertainty (YES-NO)	0.006 (0.261)		0.039 (1.121)		0.029 (1.160)		0.059 (1.446)	
Panel B. The impact of EPU on dividend payout of closely held (family) firms in countries with strong shareholder protection vs. countries with weak shareholder protection								
Level of Shareholder Protection	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Full sample				Excluding U.S. Firms			
	Firms with blockholders		Family firms		Firms with blockholders		Family firms	
	STRONG	WEAK	STRONG	WEAK	STRONG	WEAK	STRONG	WEAK
<i>EPU</i>	0.057 (1.471)	0.158*** (6.324)	−0.015 (−0.157)	0.159*** (4.956)	0.055 (1.480)	0.186*** (6.614)	0.014 (0.150)	0.232*** (4.725)
<i>SRVOL</i>	−0.407*** (−3.537)	−0.083 (−1.196)	0.189 (1.158)	−0.226** (−2.191)	−0.398*** (−3.442)	−0.037 (−0.421)	0.111 (0.667)	−0.320 (−1.392)
<i>RE</i>	0.013 (1.071)	0.002 (0.496)	0.072*** (2.932)	0.002 (0.259)	0.012 (1.069)	0.009 (1.494)	0.082*** (2.790)	0.015 (0.679)
<i>TE</i>	0.772*** (5.366)	0.135** (2.538)	0.605*** (3.174)	0.100 (0.999)	0.765*** (5.380)	0.056 (0.941)	0.610*** (3.246)	−0.277** (−2.146)
<i>ROA</i>	−0.217	−0.103*	−0.234	−0.102	−0.230	−0.269**	−0.303	−0.136

(continued on next page)

Table 13 (continued)

Panel B. The impact of EPU on dividend payout of closely held (family) firms in countries with strong shareholder protection vs. countries with weak shareholder protection								
Level of Shareholder Protection	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Full sample				Excluding U.S. Firms			
	Firms with blockholders		Family firms		Firms with blockholders		Family firms	
	STRONG	WEAK	STRONG	WEAK	STRONG	WEAK	STRONG	WEAK
<i>SGR</i>	(−0.867) −0.153***	(−1.881) −0.025***	(−0.519) −0.049	(−0.782) 0.015	(−0.918) −0.152***	(−2.491) −0.024***	(−0.678) −0.049	(−0.410) −0.004
<i>SIZE</i>	(−3.759) 0.121***	(−3.686) 0.061***	(−1.291) 0.173**	(0.696) 0.024	(−3.778) 0.122***	(−3.014) 0.051***	(−1.307) 0.169**	(−0.102) −0.140***
<i>CASH</i>	(3.030) 0.109	(3.736) −0.049	(2.300) 0.283*	(0.813) 0.158	(3.150) 0.113	(2.956) −0.105**	(2.273) 0.269*	(−2.637) −0.031
<i>GDP_PC</i>	(0.967) 0.020	(−0.954) 0.192**	(1.871) −0.086	(1.589) 0.767***	(1.003) 0.016	(−1.989) 0.270***	(1.812) −0.187	(−0.250) 0.669***
Observations	(0.074) 7415	(2.552) 13,913	(−0.190) 2483	(4.648) 6210	(0.064) 7428	(2.699) 11,179	(−0.411) 2460	(2.672) 2757
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Adj R-sq	0.296	0.435	0.404	0.320	0.293	0.459	0.390	0.380
Difference in the coefficient on EPU (STRONG – WEAK)	−0.101** (−2.180)		−0.174* (−1.704)		−0.131*** (−2.823)		−0.218** (−2.039)	

This table presents split-sample tests that examine the influence of large shareholders on the positive relation between EPU and dividends payout. The dependent variable is *PAYOUT RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). *EPU* is the natural logarithm of the average BBD (2016) policy uncertainty index over the 12-month period ending in the month of the fiscal year-end. Panel A presents the results of subsamples split by whether firms are owned by blockholders in Models 1 and 2 and Models 5 and 6, and by whether firms are owned by family in Models 3 and 4 and Models 7 and 8. In Panel B, we divide the sample into above- and below-mean subsamples according to the mean values of *SHAREHOLDER PROTECT* for a sample of firms owned by blockholders in Models 1 and 2 and Models 5 and 6, as well as for a sample of family-owned firms in Models 3 and 4 and Models 7 and 8. In both Panels A and B, Models 1 through 4 use the full sample of 19 countries, while Models 5 through 8 use the sample excluding U.S. firms. All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm-fixed effects model, with standard errors adjusted for clustering by firm. t-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 14

EPU and dividend payout: the role of board characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	INDEP_CHAIRMAN		BOARD_OWNERSHIP		BOARD_INDEP		EXTERNALBLOCK		BOARD_INDEP_EXPAND	
	YES	NO	HIGH	LOW	HIGH	LOW	HIGH	LOW	HIGH	LOW
<i>EPU</i>	−0.102 (−1.474)	0.124*** (2.625)	−0.137 (−1.643)	0.104** (2.435)	0.018 (0.312)	0.163*** (3.080)	−0.054 (−0.869)	0.167*** (3.305)	−0.030 (−1.366)	0.049** (2.370)
<i>SRVOL</i>	−0.171 (−0.567)	0.244 (1.218)	0.691* (1.678)	−0.138 (−0.787)	−0.117 (−0.621)	0.306 (1.186)	−0.397 (−1.595)	0.176 (0.892)	−0.191*** (−3.950)	−0.388*** (−4.877)
<i>RE</i>	0.016 (1.327)	0.011 (0.810)	0.002 (0.110)	0.020** (2.088)	0.007 (0.844)	0.043 (1.638)	0.009 (0.613)	0.027** (2.433)	−0.000 (−0.098)	0.014** (2.355)
<i>TE</i>	0.452*** (2.772)	0.680*** (5.388)	1.070*** (3.963)	0.355*** (3.530)	0.596*** (4.962)	0.415** (2.430)	0.368** (2.377)	0.498*** (4.380)	0.133*** (3.637)	0.185*** (3.537)
<i>ROA</i>	−0.409** (−2.167)	−0.212 (−1.294)	−0.263 (−0.676)	−0.145 (−1.189)	−0.052 (−0.468)	−0.854** (−2.440)	−0.098 (−0.545)	−0.183 (−1.351)	−0.026 (−1.318)	−0.343*** (−4.116)
<i>SGR</i>	−0.047 (−0.942)	−0.163*** (−3.578)	−0.149* (−1.854)	−0.161*** (−4.508)	−0.139*** (−3.814)	−0.150*** (−2.660)	−0.074 (−1.329)	−0.193*** (−5.460)	−0.023*** (−3.300)	−0.076*** (−4.428)
<i>SIZE</i>	0.108** (2.271)	0.057 (1.523)	0.093 (1.042)	0.089** (2.314)	0.040 (1.168)	0.110 (1.644)	−0.030 (−0.522)	0.100** (2.313)	0.020** (2.194)	0.087*** (7.012)
<i>CASH</i>	0.092 (0.540)	−0.050 (−0.427)	0.045 (0.262)	−0.004 (−0.032)	0.018 (0.175)	−0.164 (−1.019)	−0.023 (−0.130)	0.012 (0.109)	0.049* (1.660)	0.010 (0.178)
<i>GDP_PC</i>	−1.187* (−1.769)	−0.019 (−0.086)	−0.669 (−1.635)	−0.170 (−0.695)	−1.048** (−2.097)	0.005 (0.019)	−0.364 (−0.746)	−0.105 (−0.431)	−0.082 (−1.504)	−0.097* (−1.710)
Observations	3630	7166	2148	9058	6612	4594	3220	7986	28,015	21,903
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj R-sq	0.501	0.394	0.468	0.403	0.518	0.328	0.449	0.405	0.441	0.263
Difference in the coefficient on EPU (High/YES – Low/NO)	−0.226*** (−2.680)		−0.241*** (−2.578)		−0.145* (−1.846)		−0.221*** (−2.727)		−0.079*** (−2.598)	

This table presents estimation results for various subsamples of regressions of *PAYOUT RATIO* on measures of economic policy uncertainty (*EPU*) and firm- and country-level controls. The dependent variable is *PAYOUT RATIO*, defined as the ratio of cash dividends (DVC) to net income before extraordinary items (IB). *EPU* is the natural logarithm of the average BBD (2016) policy uncertainty index over the 12-month period ending in the month of the fiscal year-end. We divide the sample into above- and below-mean subsamples according to the mean values (or 1/0 values) of the partitioning variables, namely, *INDEP_CHAIRMAN* (Models 1 and 2), *BOARD_OWNERSHIP* (Models 3 and 4), *BOARD_INDEP* (Models 5 and 6), *EXTERNALBLOCK* (Models 7 and 8), and *BOARD_INDEP_EXPAND* (Models 9 and 10). All variables are defined in Table 1. Year and firm fixed effects are included in all regressions. All parameters are estimated using a firm fixed effects model, with standard errors adjusted for clustering by firm. *t*-statistics are reported beneath each coefficient estimate in parentheses. Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

increased agency costs of free cash flow, firms will pay out more dividends. We find evidence for the latter view. Specifically, we find that EPU is significantly related to various dividend payout ratios (dividend-to-earnings, dividend-to-price, dividend-to-sales, and dividend-to-cash flow), dividend per share, total payout ratio, and the cumulative abnormal return around the announcement of dividend increases. Of note, our findings are not confounded by the effects of EPU on the scaling variables (e.g., sales, net income, market capitalization) and by the effects of other sources of uncertainty, including at the firm level. The positive relation between policy uncertainty and corporate payouts is robust to including additional controls for corporate investment policy, financial soundness, uncertainty at the firm, industry, and country levels, GDP growth, election uncertainty, market expectations about future economic conditions, degree of social trust, individual investors' tax rates, and strength of investor protection. Moreover, our results are not sensitive to sample composition, more complex error structures, or to controls for endogeneity problems using multiple approaches, including a two-stage instrumental variables analysis and a placebo test.

In additional analyses, we find further evidence in support of the agency hypothesis. In particular, the effect of policy uncertainty on dividend payouts is more pronounced in firms with fewer growth opportunities and positive free cash flows, in firms located in countries with weak shareholder protection, in closely held/family firms when shareholder protection is weak, and in firms with weak internal board governance. We also find that shareholder pressure to extract cash is attenuated in countries with weak creditor protection, where managers use a more conservative dividend policy as a substitute method to reduce agency conflicts between shareholders and creditors. Viewed from the lens of agency theory, our novel evidence suggests that firms pay dividends to mitigate agency problems during high-EPU periods.

Our novel evidence identifies EPU as a missing factor that can help explain firm payout policy. An interesting finding revealed by our study is that the effect of EPU on dividends is distinct from that of firm-level uncertainty and from other sources of non-firm uncertainty, which may be attributed to different perceived potential agency problems (and their severity) with different sources of uncertainty. These considerations offer promising directions for future research. In addition, our study suggests that the effect of EPU on payout is more pronounced on cash dividends than on buybacks. This is seemingly puzzling, given the general stickiness of dividends and the ability of share repurchases to preserve a firm's financial flexibility. Future theoretical research along these lines is warranted. Future studies may also extend our analysis by examining the interactions among dividend policies and other corporate policies during high-EPU periods.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcorpfin.2020.101785>.

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